

# **BLUESTOCK MUTUAL FUND ANALYTICS CAPSTONE**

## **Final Project Report**

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**Bluestock Fintech Capstone Project**

**Mutual Fund Analytics Platform**

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## 1. EXECUTIVE SUMMARY

The Mutual Fund Analytics Capstone Project was designed to develop a comprehensive analytics framework capable of transforming raw mutual fund datasets into meaningful business insights. The project demonstrates the practical application of data engineering, exploratory analysis, financial performance evaluation, risk analytics, and business intelligence visualization within the financial services domain.

The primary objective of the project was to analyze mutual fund data from multiple perspectives including fund performance, investor participation, portfolio composition, and risk-adjusted returns. To achieve this objective, an end-to-end workflow was implemented consisting of data extraction, transformation, loading (ETL), exploratory data analysis (EDA), performance analytics, risk assessment, and dashboard development.

Multiple datasets including Net Asset Value (NAV) history, Assets Under Management (AUM), Systematic Investment Plan (SIP) inflows, investor demographics, and portfolio holdings were analyzed using Python-based analytical tools. Data cleaning and preprocessing activities ensured consistency and reliability across datasets.

Several performance metrics including CAGR, Annualized Return, Volatility, Sharpe Ratio, Sortino Ratio, and Maximum Drawdown were calculated to evaluate mutual fund performance. Additionally, correlation analysis and risk-return assessments were performed to provide deeper insights into investment behavior and fund characteristics.

An interactive Power BI dashboard was developed to present analytical results through dynamic visualizations and business-friendly reporting interfaces. The dashboard enables users to evaluate performance trends, assess risk exposure, monitor investor activity, and compare mutual fund performance efficiently.

The project successfully demonstrates how modern data analytics and business intelligence methodologies can improve investment analysis and support informed decision-making in the mutual fund industry.

## 2. PROJECT OVERVIEW

Mutual funds have become one of the most widely used investment instruments due to their diversification benefits and professional management. However, evaluating mutual fund performance requires analysis across multiple dimensions such as returns, risk, investor behavior, and portfolio composition.

Traditional reporting approaches often present large volumes of data without sufficient analytical context, making it difficult for investors and analysts to derive actionable insights. This project addresses this challenge by creating an integrated analytics platform capable of processing, analyzing, and visualizing mutual fund data.

The project follows a structured workflow beginning with data acquisition and preprocessing, followed by exploratory analysis, performance measurement, risk assessment, and dashboard reporting. The resulting solution provides stakeholders with a centralized platform for evaluating mutual fund performance and investment trends.

Project Objectives:

- Build an automated ETL pipeline.
- Perform exploratory data analysis.
- Calculate performance metrics.
- Conduct risk analytics.
- Analyze investor demographics.
- Develop an interactive dashboard.
- Generate actionable business insights.

The project demonstrates practical implementation of data analytics techniques within the financial services sector.

The project successfully demonstrates how modern data analytics and business intelligence methodologies can improve investment analysis and support informed decision-making in the mutual fund industry.

**Project Repository (GitHub):**

[https://github.com/Pardhu9114/bluestock\\_mf\\_capstone](https://github.com/Pardhu9114/bluestock_mf_capstone)

### **3. DATA SOURCES**

The Mutual Fund Analytics Capstone Project utilizes multiple datasets representing different dimensions of the mutual fund ecosystem. These datasets collectively provide information regarding fund performance, investor participation, asset growth, portfolio composition, and benchmark performance. The integration of these datasets enables comprehensive financial analysis and business intelligence reporting.

#### **3.1 Fund Master Dataset**

The Fund Master dataset serves as the primary reference table containing detailed information about mutual fund schemes. It includes scheme identifiers, fund names, categories, fund houses, and AMFI codes. This dataset was used to establish relationships between various datasets and ensure consistency throughout the analysis process.

Purpose:

- Scheme identification
  - Fund categorization
  - Data integration and mapping
  - Reference information management
- 

#### **3.2 NAV History Dataset**

The NAV (Net Asset Value) History dataset contains historical daily NAV values for multiple mutual fund schemes. NAV represents the per-unit value of a mutual fund and serves as a critical indicator of fund performance over time.

This dataset was extensively used to perform trend analysis, calculate returns, evaluate fund growth, and derive risk-adjusted performance metrics.

Purpose:

- Performance measurement
- Return calculations
- Trend analysis

- Volatility estimation
  - Risk analytics
- 

### **3.3 Assets Under Management (AUM) Dataset**

The AUM dataset contains information regarding the total assets managed by different mutual fund houses. Assets Under Management represent the cumulative market value of investments handled by a fund manager.

The dataset was used to assess fund house growth, investor confidence, and market participation trends.

Purpose:

- Fund size comparison
  - Growth analysis
  - Market share evaluation
  - Investor confidence assessment
- 

### **3.4 Monthly SIP Inflows Dataset**

The Monthly SIP Inflows dataset captures the total value of investments received through Systematic Investment Plans (SIPs) on a monthly basis.

SIP inflows serve as an important indicator of retail investor participation and long-term investment behavior. This dataset was analyzed to identify growth trends and investor commitment towards mutual fund investments.

Purpose:

- Investor participation analysis
- SIP growth tracking
- Long-term investment behavior assessment
- Market sentiment evaluation

### **3.5 Category Inflows Dataset**

The Category Inflows dataset provides category-wise investment inflows across different mutual fund segments such as Equity, Debt, Hybrid, and Solution-Oriented schemes.

This dataset was used to understand investor preferences and identify the categories attracting the highest capital inflows.

Purpose:

- Category performance analysis
  - Investment preference assessment
  - Market trend identification
  - Capital flow analysis
- 

### **3.6 Industry Folio Dataset**

The Industry Folio dataset contains information regarding the number of investor folios across the mutual fund industry.

A folio represents an individual investor account within a mutual fund scheme. Growth in folio counts is often considered an indicator of increasing retail participation and market penetration.

Purpose:

- Investor participation analysis
  - Industry growth assessment
  - Market adoption measurement
  - Retail investor tracking
- 

### **3.7 Investor Transactions Dataset**

The Investor Transactions dataset contains transaction-level information related to investor activities such as purchases, redemptions, and SIP contributions.

This dataset was utilized to study investment patterns, transaction behavior, and demographic participation trends.

Purpose:

- Investor behavior analysis
  - Transaction pattern evaluation
  - Demographic analysis
  - Investment activity assessment
- 

### **3.8 Portfolio Holdings Dataset**

The Portfolio Holdings dataset contains detailed information regarding the securities and sectors held within mutual fund portfolios.

The dataset enables portfolio composition analysis, sector allocation studies, and diversification assessments.

Purpose:

- Sector allocation analysis
- Portfolio diversification evaluation
- Risk exposure assessment
- Holdings concentration analysis

### **3.9 Benchmark Indices Dataset**

The Benchmark Indices dataset contains historical benchmark performance data used to compare mutual fund returns against market indices.

Benchmark analysis provides context for evaluating whether mutual funds are outperforming or underperforming the broader market.

Purpose:

- Benchmark comparison
- Relative performance analysis
- Alpha generation assessment
- Market performance evaluation

## 4. ETL DESIGN AND METHODOLOGY

### 4.1 Overview

The ETL (Extract, Transform, Load) process forms the foundation of the Mutual Fund Analytics Platform developed in this project. ETL is a critical data engineering methodology used to collect data from multiple sources, improve data quality through preprocessing and validation, and prepare structured datasets for analytics and visualization.

Given the diverse nature of mutual fund datasets, including NAV history, AUM information, SIP inflows, investor transactions, portfolio holdings, and benchmark performance data, a robust ETL pipeline was required to ensure consistency, accuracy, and reliability. The ETL workflow was implemented using Python and the Pandas library, enabling automated processing of large datasets while minimizing manual intervention.

The overall workflow begins with raw CSV datasets and progresses through data ingestion, cleaning, validation, transformation, storage, and reporting stages before ultimately supporting dashboard visualization in Power BI. The ETL architecture was designed to be modular, scalable, and reusable, allowing future integration of additional datasets and automated refresh processes.

(Insert Figure 1: ETL Workflow Diagram Here)

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### 4.2 Extract Phase

The extraction phase focused on acquiring data from multiple structured CSV files representing various aspects of the mutual fund industry. These datasets included information related to fund schemes, historical NAV values, assets under management, SIP inflows, investor transactions, portfolio holdings, category-wise fund flows, folio counts, and benchmark indices.

Python's Pandas library was utilized to read and import the datasets into DataFrame structures for further processing. During extraction, initial validation checks were performed to ensure that all required files were available and contained the expected schema.

The following activities were carried out during the extraction stage:

- Importing raw CSV datasets into Pandas DataFrames.
- Verifying file availability and structure.
- Inspecting dataset dimensions and attributes.

- Validating column names and schema consistency.
- Performing preliminary quality assessments.

The extraction phase established the foundation for downstream processing by ensuring that data from all sources could be successfully integrated into the analytics workflow.

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### **4.3 Transform Phase**

The transformation phase was the most significant stage of the ETL process. Raw financial datasets often contain inconsistencies, duplicate records, missing values, and formatting issues that can negatively impact analytical accuracy. Therefore, extensive preprocessing and data quality improvements were implemented.

Several transformation techniques were applied to prepare the datasets for analysis:

#### **Data Cleaning**

Data cleaning involved identifying and correcting inconsistencies within the datasets. Duplicate records were removed, missing values were handled appropriately, and invalid entries were corrected wherever necessary.

#### **Data Standardization**

Date fields were converted into standardized formats to support time-series analysis and ensure consistency across datasets. Numerical fields were validated and converted into appropriate data types for accurate calculations.

#### **Data Validation**

Data validation checks were performed to verify the integrity of the datasets. These checks ensured that numerical values fell within expected ranges and that relationships between datasets remained consistent.

#### **Feature Engineering**

Additional derived attributes were created to support advanced analytics. Examples include return calculations, growth metrics, and risk indicators that were later used for performance and risk analysis.

#### **Consistency Verification**

Cross-dataset validation was performed to ensure that scheme identifiers, categories, and benchmark references aligned correctly across all data sources.

The transformation phase significantly improved the quality, reliability, and analytical readiness of the data, enabling accurate reporting and visualization.

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#### **4.4 Load Phase**

After data cleaning and transformation, the processed datasets were loaded into structured storage formats for further analysis and reporting. The cleaned data was organized systematically and stored within the project environment to facilitate efficient access by analytics scripts and dashboard components.

The project utilized structured storage mechanisms including processed datasets and a centralized SQLite database. This approach enabled efficient querying, data retrieval, and integration with reporting tools.

The loading phase involved:

- Saving processed datasets for analytical use.
- Creating structured tables within the SQLite database.
- Maintaining consistency across processed data assets.
- Supporting integration with visualization and reporting tools.

The SQLite database acted as a centralized repository, ensuring efficient access to validated and transformed data throughout the project lifecycle.

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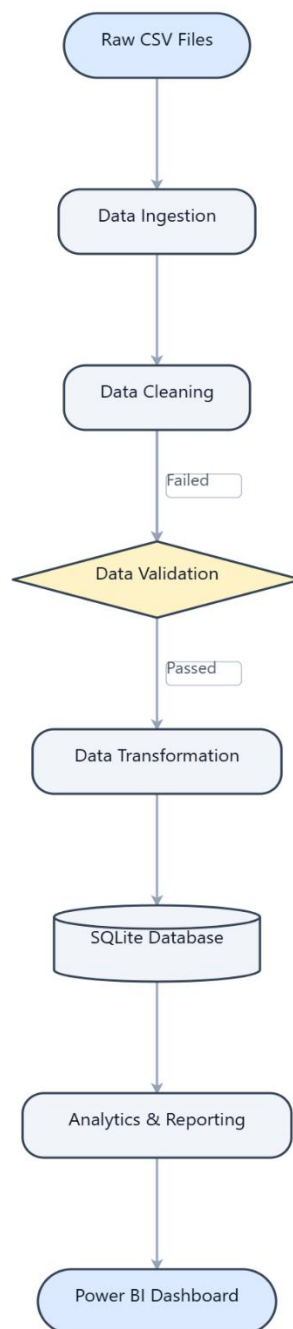
#### **4.5 Benefits of the ETL Pipeline**

The implementation of a structured ETL pipeline provided several important benefits to the project.

- Improved data quality and reliability.
- Automated repetitive preprocessing tasks.
- Reduced manual intervention and human error.
- Enhanced consistency across multiple datasets.

- Enabled scalable analytics workflows.
- Facilitated efficient integration with Power BI dashboards.
- Supported reproducible and maintainable data processing.

By establishing a robust ETL framework, the project created a reliable foundation for exploratory analysis, performance evaluation, risk assessment, and business intelligence reporting. The ETL pipeline ensured that all subsequent analytical insights were derived from accurate, validated, and standardized data sources.



**Figure 1: End-to-End ETL Workflow for the Mutual Fund Analytics Platform**

## 5. EXPLORATORY DATA ANALYSIS (EDA) FINDINGS

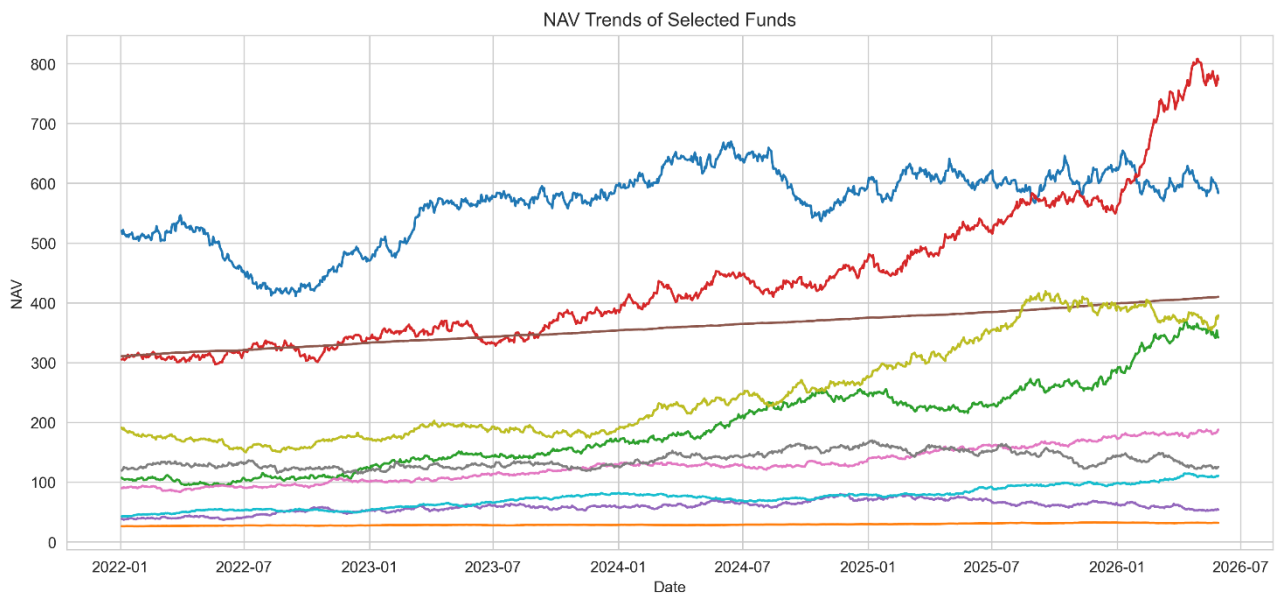
### 5.1 NAV Trend Analysis

Net Asset Value (NAV) is one of the most important indicators of mutual fund performance as it reflects the per-unit market value of a fund. Historical NAV data was analyzed to understand growth trends, market behavior, and long-term wealth creation potential across mutual fund schemes.

The NAV trend analysis revealed a generally positive growth trajectory across the observation period. Although short-term fluctuations were observed due to changing market conditions and economic events, the overall direction remained upward, indicating long-term appreciation in fund value.

The analysis highlights the effectiveness of long-term investing, where temporary market volatility is often outweighed by sustained growth over extended periods. The increasing NAV values also demonstrate the impact of compounding returns in wealth creation.

Figure 2: Historical NAV Trend Analysis



#### Key Findings

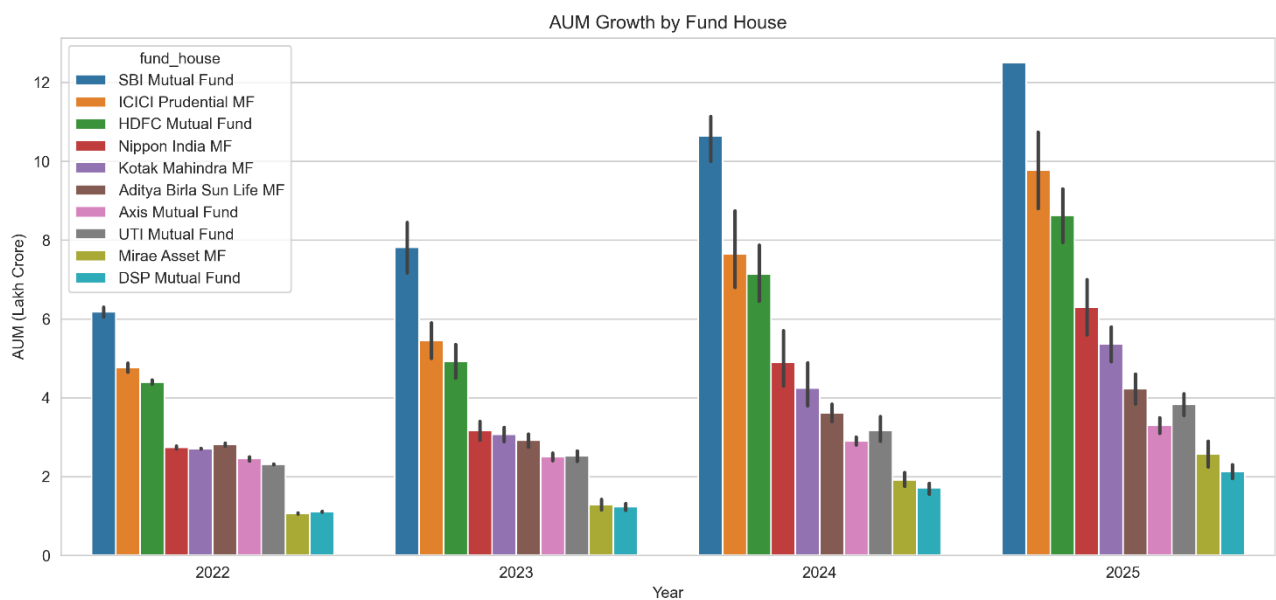
- Consistent long-term growth was observed across the majority of schemes.
- Short-term fluctuations corresponded with broader market volatility.
- Compounding effects significantly contributed to portfolio appreciation.
- Long-term investors benefited from sustained NAV growth despite temporary corrections.

Assets Under Management (AUM) represents the total market value of investments managed by a fund house. AUM is widely regarded as an indicator of investor confidence, fund popularity, and overall market participation.

The analysis of AUM data revealed substantial growth across major fund houses over the study period. Leading asset management companies continued to attract significant capital inflows, resulting in a steady expansion of managed assets.

The increasing AUM levels suggest growing investor trust in mutual fund investments and reflect the expansion of India's mutual fund industry.

Figure 3: AUM Growth by Fund House



### Key Findings

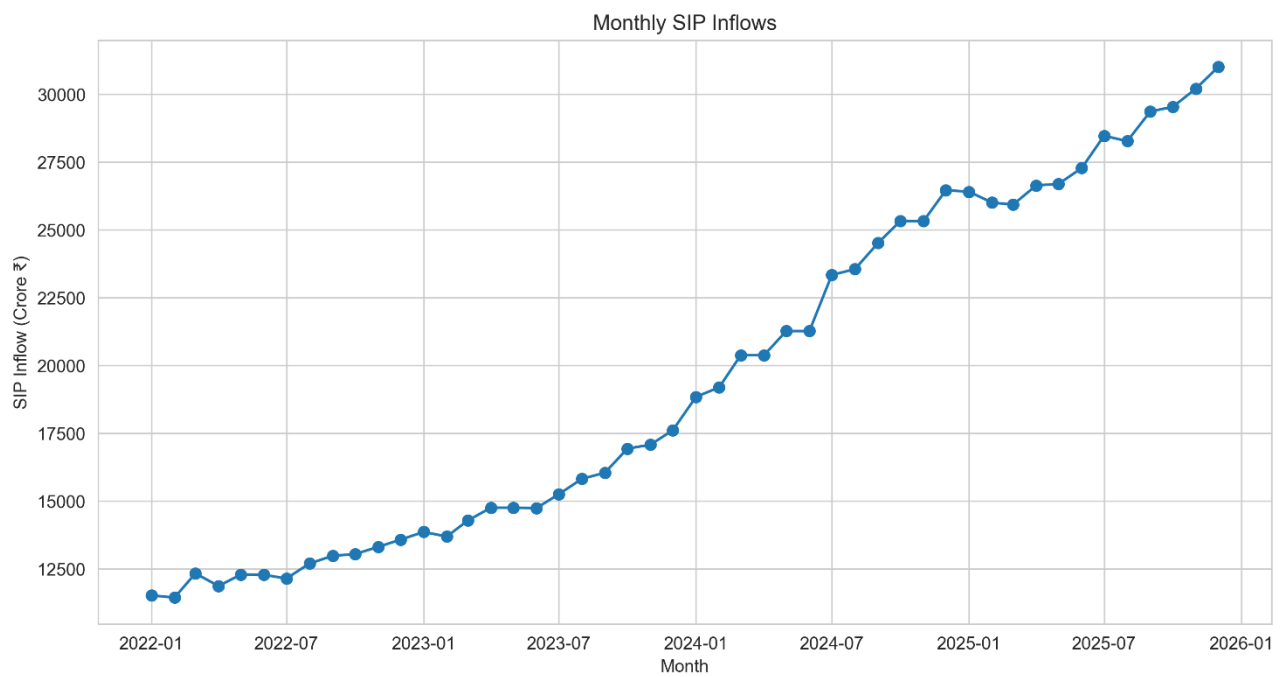
- Major fund houses demonstrated continuous AUM growth.
- Investor confidence increased significantly over time.
- Higher AUM levels indicate stronger market participation.
- Asset growth reflects increasing adoption of mutual fund investments.

### 5.3 Monthly SIP Inflow Analysis

Systematic Investment Plans (SIPs) are one of the most popular investment methods among retail investors due to their disciplined and long-term nature. Monthly SIP inflows provide valuable insights into investor sentiment and participation levels.

The analysis revealed a strong upward trend in SIP contributions, indicating increasing investor commitment towards systematic investing. The growth in SIP inflows reflects rising financial awareness and confidence in mutual funds as a long-term wealth creation vehicle.

Figure 4: Monthly SIP Inflows Trend



#### Key Findings

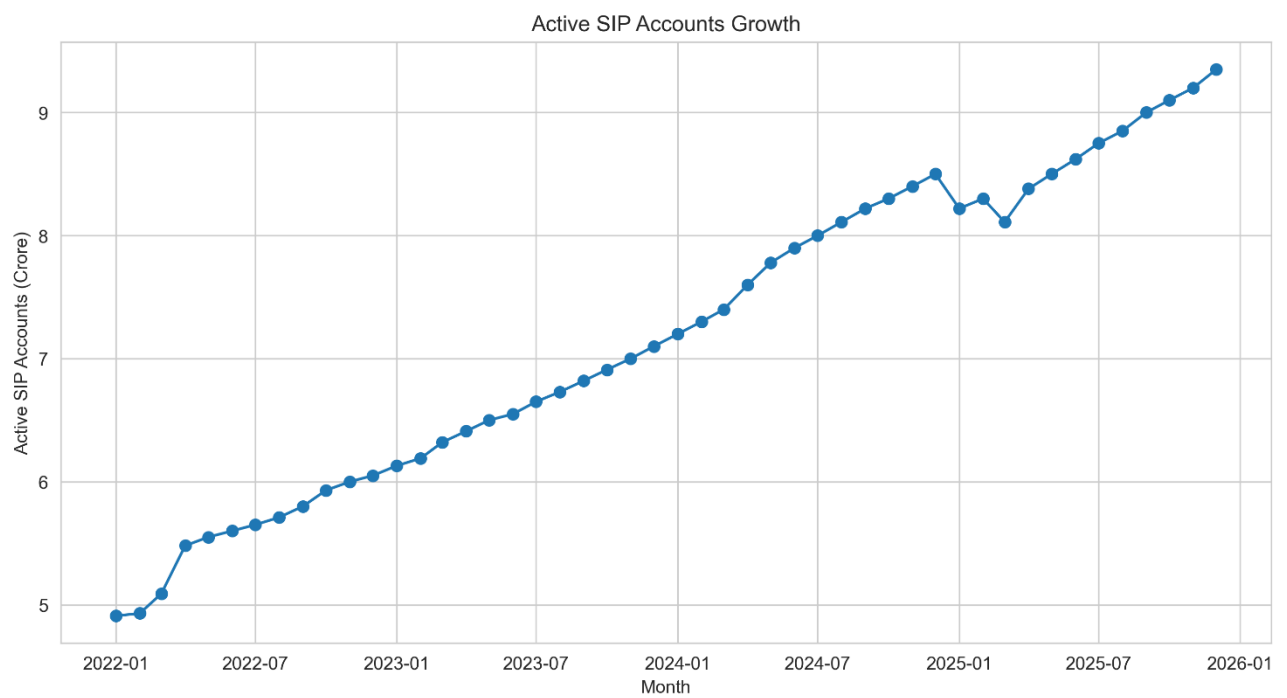
- SIP inflows increased steadily throughout the analysis period.
  - Retail investor participation continued to expand.
  - Systematic investing remained a preferred investment approach.
  - Consistent SIP contributions indicate positive market sentiment.
-

### 5.4 Active SIP Accounts Growth

Active SIP accounts represent the number of investors currently participating in systematic investment plans. Growth in active SIP accounts is a strong indicator of increasing market penetration and investor engagement.

The analysis demonstrated continuous growth in active SIP registrations, suggesting that more investors are adopting disciplined investment practices and participating in mutual fund markets.

Figure 5: Active SIP Accounts Growth



#### Key Findings

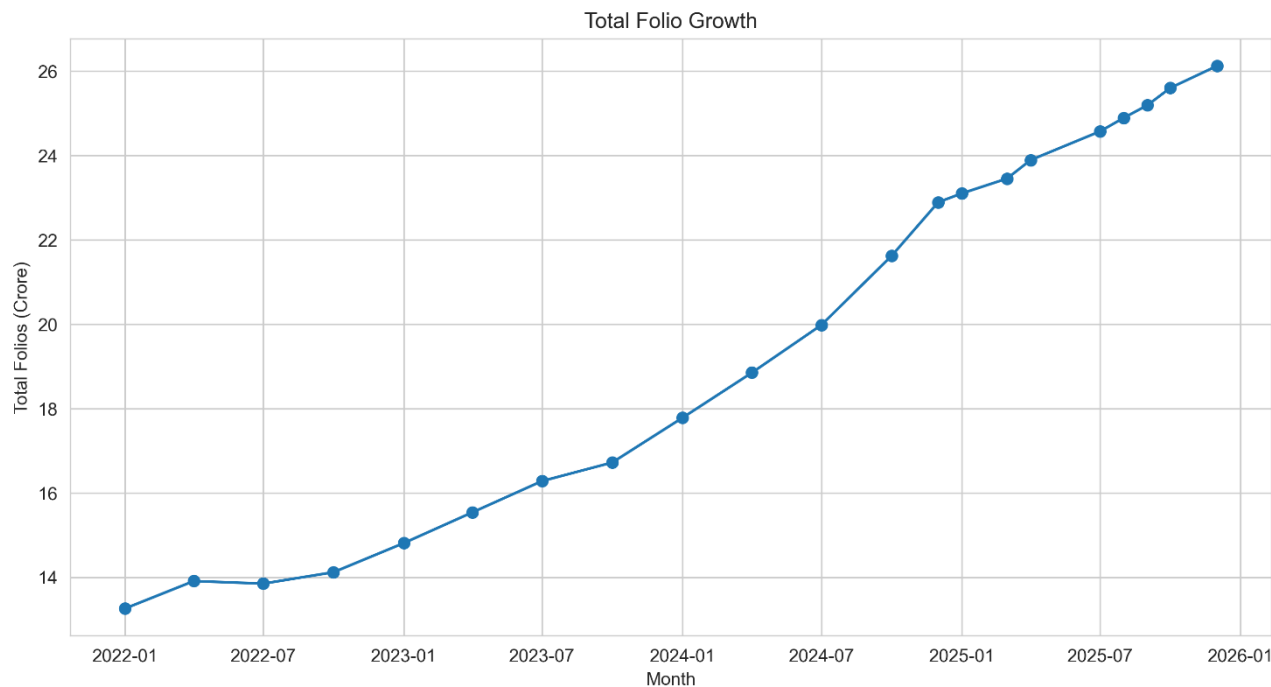
- Active SIP registrations increased consistently.
  - Retail participation expanded significantly.
  - Investors increasingly preferred recurring investment strategies.
  - Growth reflects rising awareness of long-term investing.
-

### 5.5 Industry Folio Growth Analysis

A folio represents an individual investor account within a mutual fund scheme. The number of folios serves as an important indicator of market penetration and investor participation.

The folio growth analysis showed a steady increase in investor accounts across the mutual fund industry. This trend indicates broader adoption of mutual fund investments among retail investors.

Figure 6: Total Folio Growth Trend



#### Key Findings

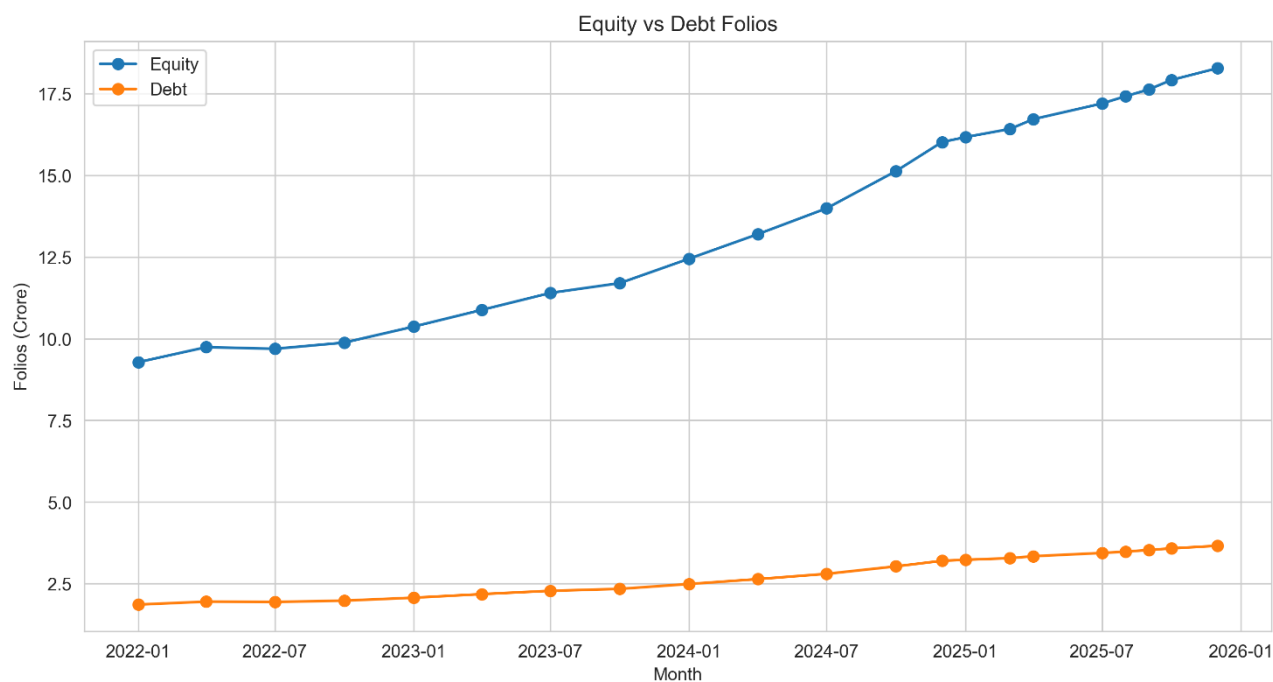
- Total folio count increased consistently over time.
  - More investors entered mutual fund markets.
  - Industry growth was supported by expanding investor participation.
  - Rising folio counts indicate growing trust in mutual fund products.
-

### 5.6 Equity vs Debt Folio Analysis

Mutual fund investors often choose between equity-oriented and debt-oriented investment products depending on their risk tolerance and investment objectives.

The analysis revealed that equity-oriented schemes accounted for a significantly larger share of investor folios compared to debt schemes. This trend reflects investor preference for long-term capital appreciation and growth-oriented investments.

Figure 7: Equity vs Debt Folios Comparison



#### Key Findings

- Equity schemes attracted significantly higher participation.
  - Investors demonstrated greater interest in growth-focused products.
  - Debt schemes maintained stable but comparatively lower participation levels.
  - Equity funds remained the dominant category within the industry.
-

5.7 Investor Demographic Analysis

Investor demographic analysis was conducted to understand participation patterns across different investor groups. Age distribution, gender participation, and investment amount patterns were examined to identify behavioural trends.

The results indicate that younger and middle-aged investors represent a significant portion of mutual fund participants. The 26–35 age group contributed the highest investment volumes, highlighting strong engagement from early-career and mid-career investors.

Figure 8: Age Group Distribution

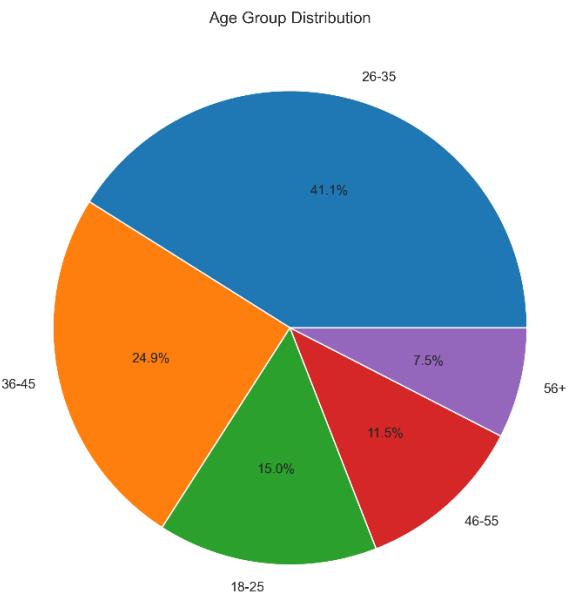


Figure 9: Gender Distribution

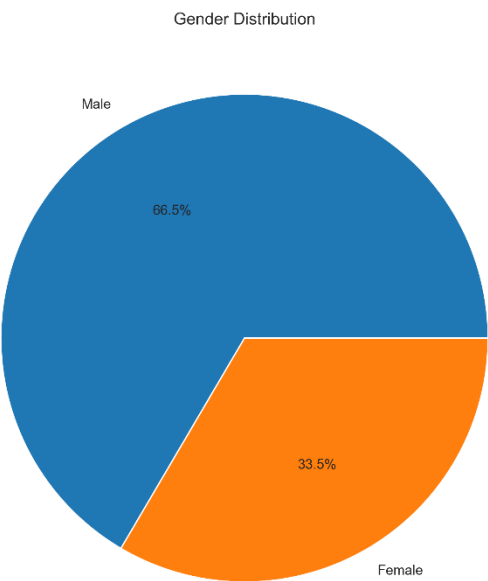
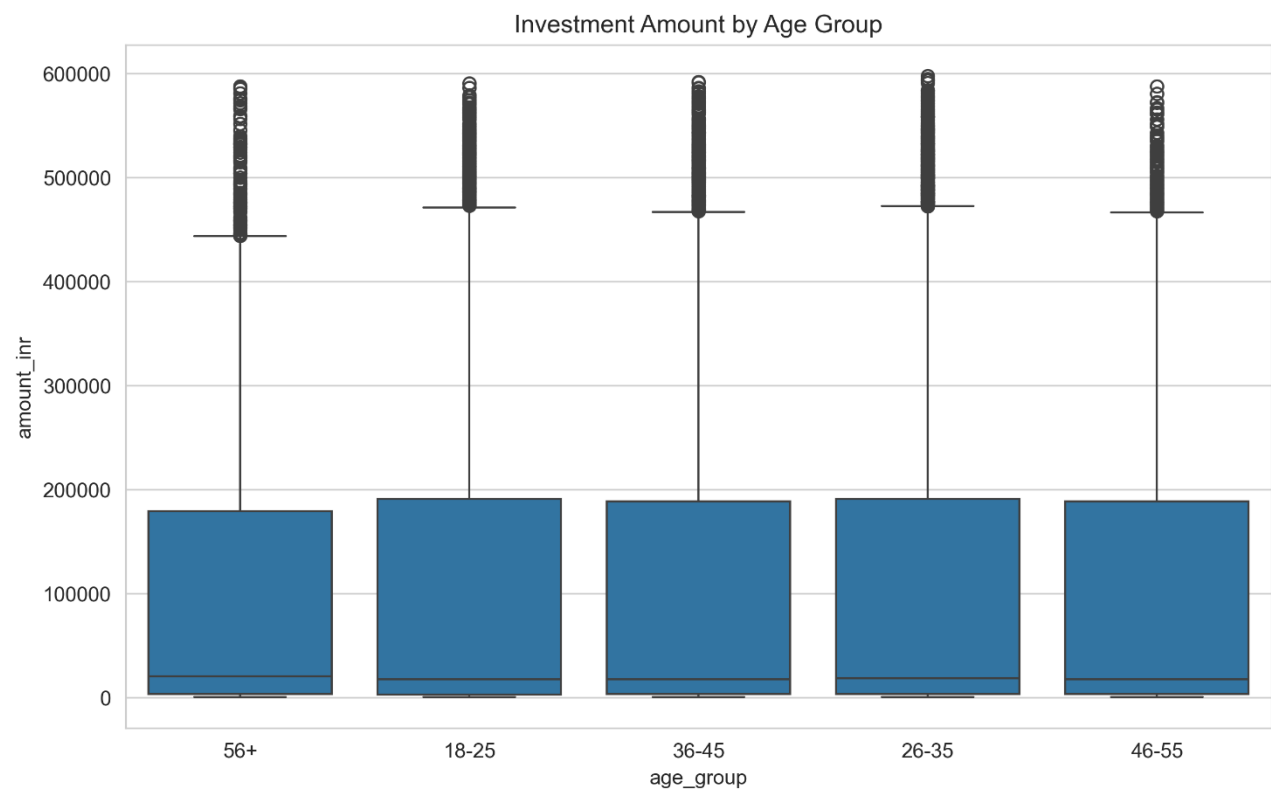


Figure 10: Investment Amount by Age Group



### Key Findings

- The 26–35 age group contributed the largest investment share.
- Younger investors increasingly participated in SIP investments.
- Investment behavior varied across demographic categories.
- A diverse investor base supports long-term industry growth.

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## 5.8 Geographic Participation Analysis

Geographic analysis was performed to understand investment activity across different regions and city categories.

The analysis revealed that T30 cities continue to contribute the majority of SIP investments. However, participation from B30 cities has also increased steadily, indicating growing market penetration beyond major metropolitan areas.

Figure 11: State-Wise SIP Contribution

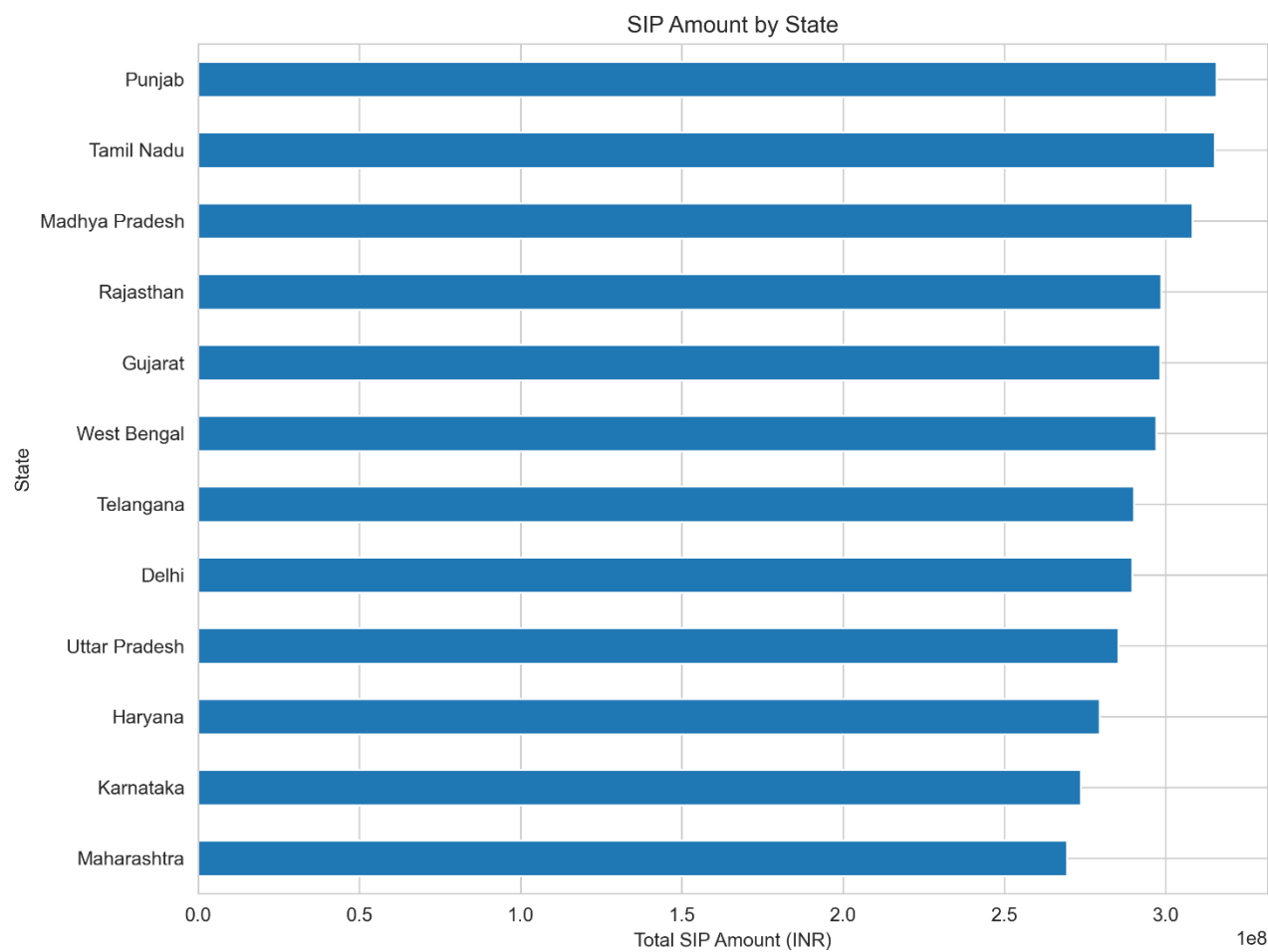
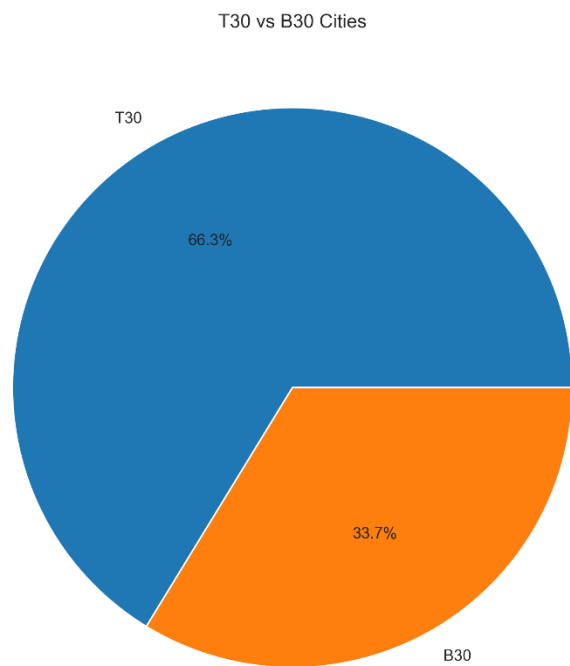


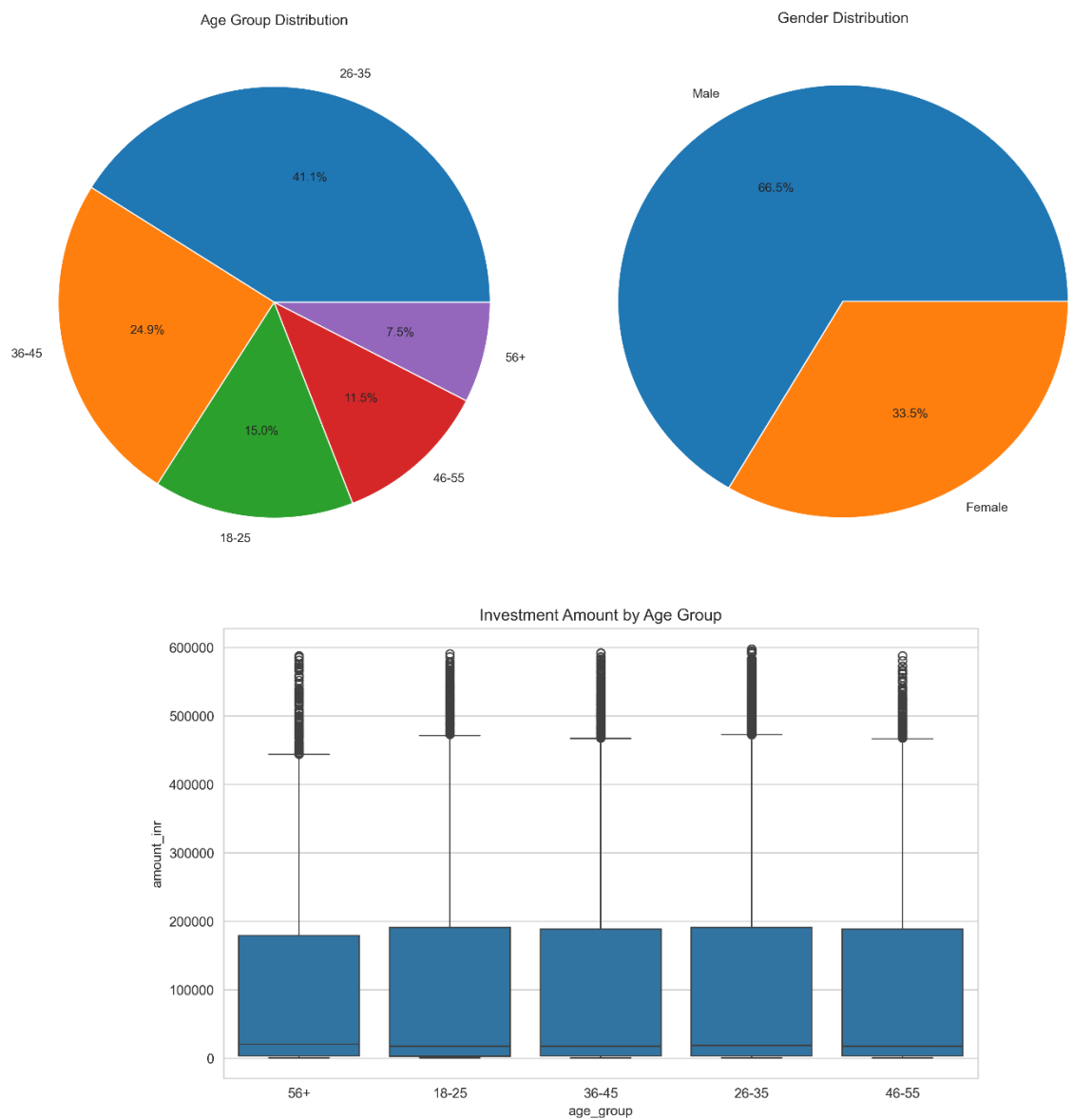
Figure 12: T30 vs B30 City Participation



Key Findings

- T30 cities account for the majority of mutual fund investments.
- B30 cities show significant growth potential.
- Urban investors remain the primary contributors to SIP inflows.
- Geographic diversification of investors continues to improve.

The EDA phase provided a comprehensive understanding of mutual fund performance trends, investor participation patterns, demographic characteristics, and market growth dynamics. These insights formed the foundation for the subsequent performance analysis and risk assessment stages of the project.



## 6. PERFORMANCE ANALYSIS

Performance analysis was conducted to evaluate the effectiveness of mutual fund investments using industry-standard financial metrics. While absolute returns provide an indication of growth, a comprehensive evaluation requires examining both return generation and associated risk levels. Therefore, multiple performance measures were calculated to assess fund efficiency, consistency, and long-term investment potential.

The analysis utilized historical NAV data and benchmark information to derive performance metrics that facilitate meaningful comparisons across mutual fund schemes.

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### 6.1 Compound Annual Growth Rate (CAGR) Analysis

Compound Annual Growth Rate (CAGR) is one of the most widely used performance metrics in the investment industry. CAGR measures the average annual growth rate of an investment over a specified period while accounting for the effects of compounding.

Unlike simple return calculations, CAGR provides a standardized measure of long-term growth and enables fair comparison between investment schemes with different performance trajectories.

The CAGR analysis conducted in this project revealed noticeable differences in long-term growth rates across mutual fund schemes. Certain funds consistently generated superior annualized returns, indicating stronger performance and effective portfolio management.

Figure 13:Top Mutual Funds Ranked by CAGR (%)

Top 10 Funds by CAGR (%)

| amfi_code | cagr  |
|-----------|-------|
| 120505.0  | 32.83 |
| 119598.0  | 32.42 |
| 149324.0  | 32.29 |
| 148569.0  | 31.95 |
| 148567.0  | 30.97 |
| 120843.0  | 30.91 |
| 100033.0  | 30.12 |
| 149323.0  | 29.58 |
| 119094.0  | 28.21 |
| 119551.0  | 25.8  |

### Key Observations

- CAGR values varied significantly across mutual fund schemes.
  - Top-performing schemes generated higher long-term compounded returns.
  - CAGR effectively highlighted the impact of long-term investing.
  - Compounding contributed substantially to wealth creation.
- 

## 6.2 Annualized Return Analysis

Annualized Return converts investment performance into a yearly growth rate, allowing performance comparison across multiple investment horizons.

This metric provides investors with a standardized benchmark for evaluating fund performance regardless of the investment duration. Annualized return calculations confirmed the presence of varying growth patterns among different mutual fund categories.

The analysis demonstrated that while some schemes generated strong annualized returns, others delivered more moderate but stable growth.

### Key Observations

- Annualized returns differed across schemes and categories.
  - Growth-oriented funds generally produced higher returns.
  - Stable funds exhibited lower but more consistent performance.
  - Long-term investment horizons improved return consistency.
- 

## 6.3 Volatility Analysis

Volatility measures the degree of variation in investment returns and serves as a primary indicator of investment risk. Higher volatility suggests greater uncertainty, while lower volatility indicates relatively stable performance.

The volatility analysis revealed differences in risk exposure across mutual fund schemes. Certain funds experienced larger fluctuations due to market sensitivity, while diversified portfolios demonstrated more stable behavior.

### Key Observations

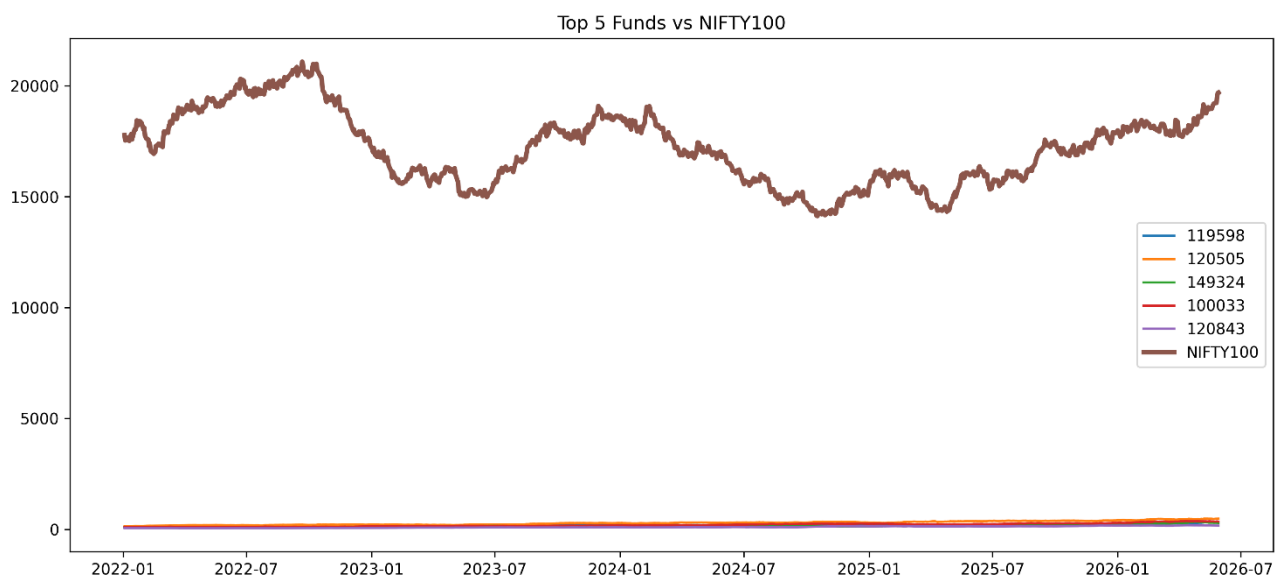
- Volatility levels varied significantly across funds.
  - Diversified portfolios generally exhibited lower volatility.
  - Higher returns were often accompanied by increased risk exposure.
  - Volatility analysis highlighted the importance of risk management.
- 

## 6.4 Benchmark Performance Comparison

Benchmark comparison is essential for evaluating whether a mutual fund has generated returns beyond what could be achieved through passive market exposure.

Benchmark indices were used as reference points to assess fund performance relative to broader market movements. The comparison revealed that several mutual fund schemes outperformed their respective benchmarks, while others closely tracked market performance.

Figure 14: Top 5 Mutual Funds Compared with NIFTY100 Benchmark



### Key Observations

- Several schemes outperformed benchmark indices.
  - Benchmark comparison provided context for performance evaluation.
  - Consistent outperformance indicates effective fund management.
  - Relative performance assessment improved investment decision-making.
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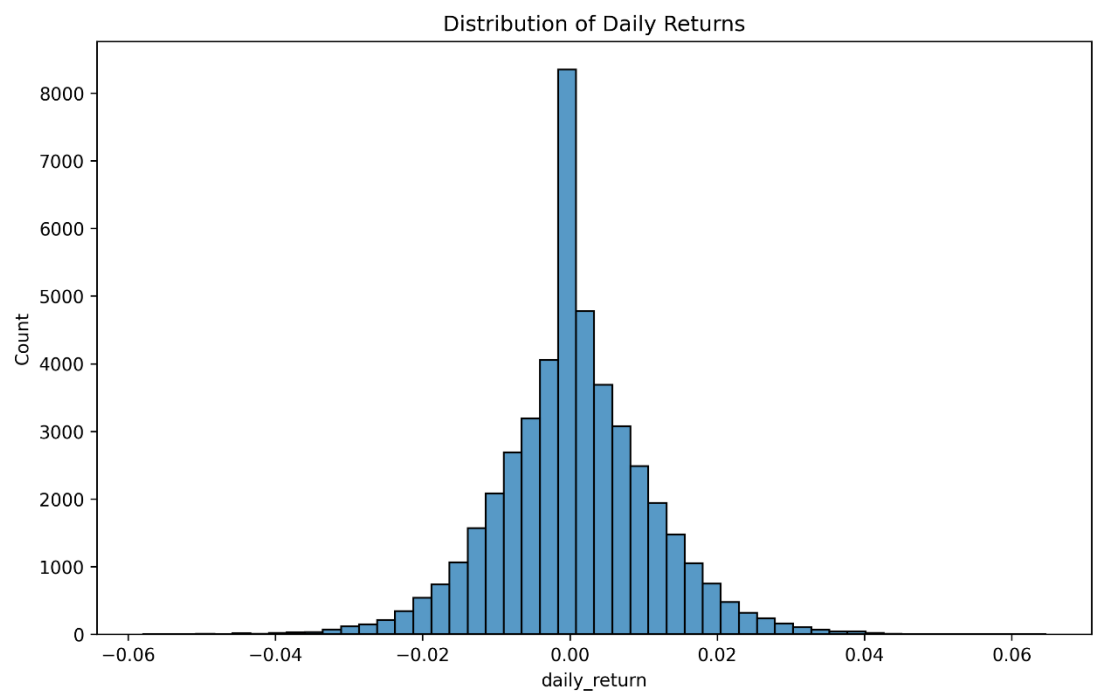
### 6.5 Daily Return Distribution Analysis

Daily return analysis was performed to understand the distribution and consistency of returns generated by mutual fund schemes.

The distribution patterns indicated that most returns remained concentrated around average values, while extreme positive or negative return events occurred less frequently.

Understanding return distribution is important because it provides insights into risk behavior and helps investors evaluate the likelihood of unexpected performance fluctuations.

Figure 15: Distribution of Daily Mutual Fund Returns



#### Key Observations

- Most returns clustered around the mean return value.
- Extreme return events were relatively infrequent.
- Distribution analysis improved understanding of investment risk.
- Return consistency varied across schemes.

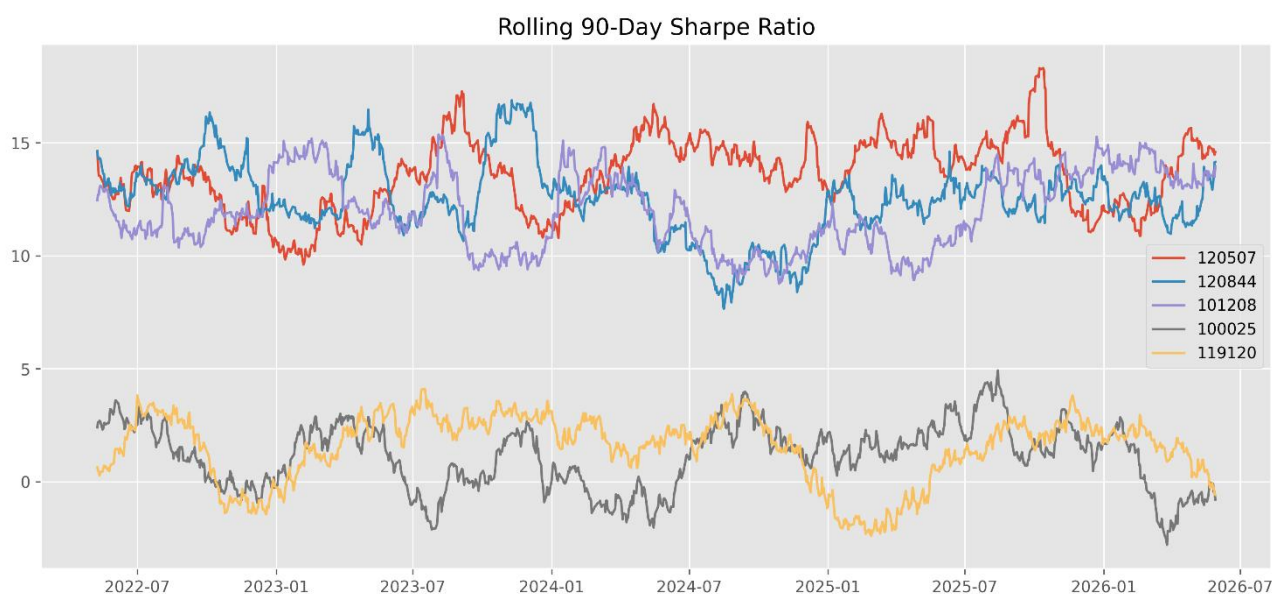
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### 6.6 Sharpe Ratio Analysis

The Sharpe Ratio is one of the most widely used measures of risk-adjusted performance. It evaluates how much excess return is generated for each unit of risk undertaken.

A higher Sharpe Ratio indicates greater efficiency in converting risk exposure into investment returns. The analysis revealed significant variation in Sharpe Ratios across mutual fund schemes, suggesting differences in portfolio efficiency and risk management effectiveness.

**Figure 16: Top Mutual Funds Ranked by Sharpe Ratio**



#### Key Observations

- Risk-adjusted performance varied across funds.
- Higher Sharpe Ratios indicated superior investment efficiency.
- Certain funds consistently maintained stronger risk-adjusted returns.
- Rolling Sharpe analysis revealed performance stability over time.

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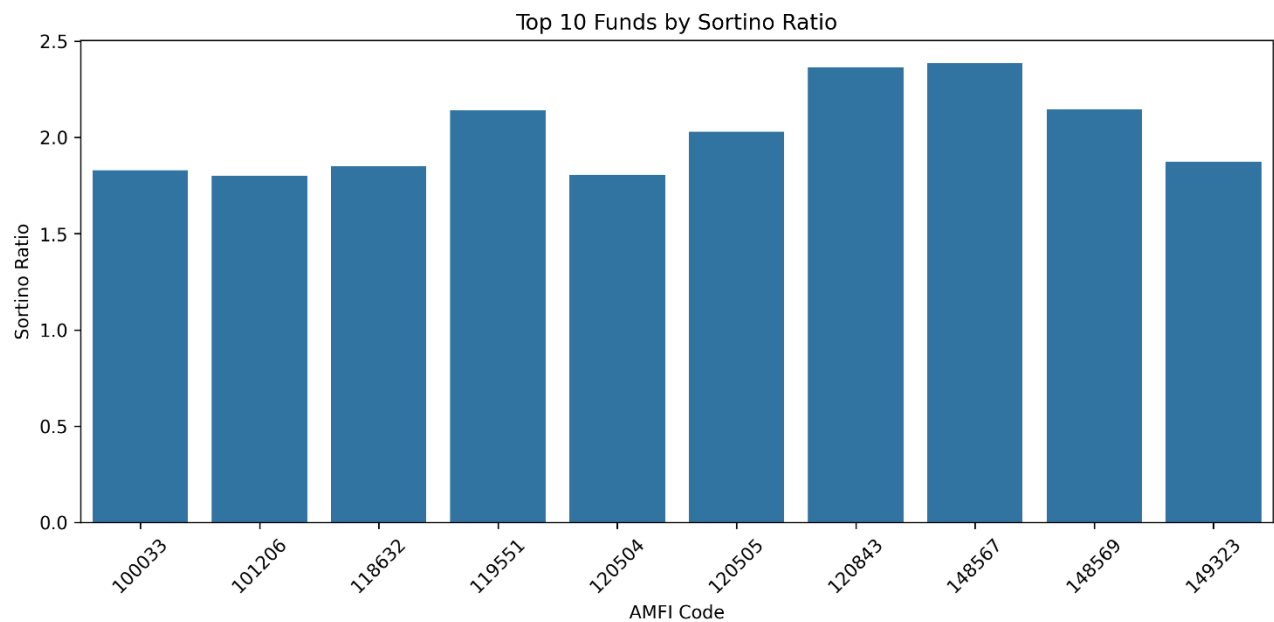
### 6.7 Sortino Ratio Analysis

While the Sharpe Ratio considers total volatility, the Sortino Ratio focuses specifically on downside risk by penalizing only negative return deviations.

This metric provides a more investor-centric perspective because investors are generally more concerned about losses than positive fluctuations.

The Sortino Ratio analysis identified funds that effectively managed downside risk while maintaining competitive returns.

Figure 17: Top Mutual Funds Ranked by Sortino Ratio



Key Observations

- Downside risk varied across schemes.
- Certain funds demonstrated superior capital preservation.
- Sortino Ratios provided additional insight beyond traditional volatility measures.
- Funds with higher Sortino Ratios exhibited stronger downside protection.

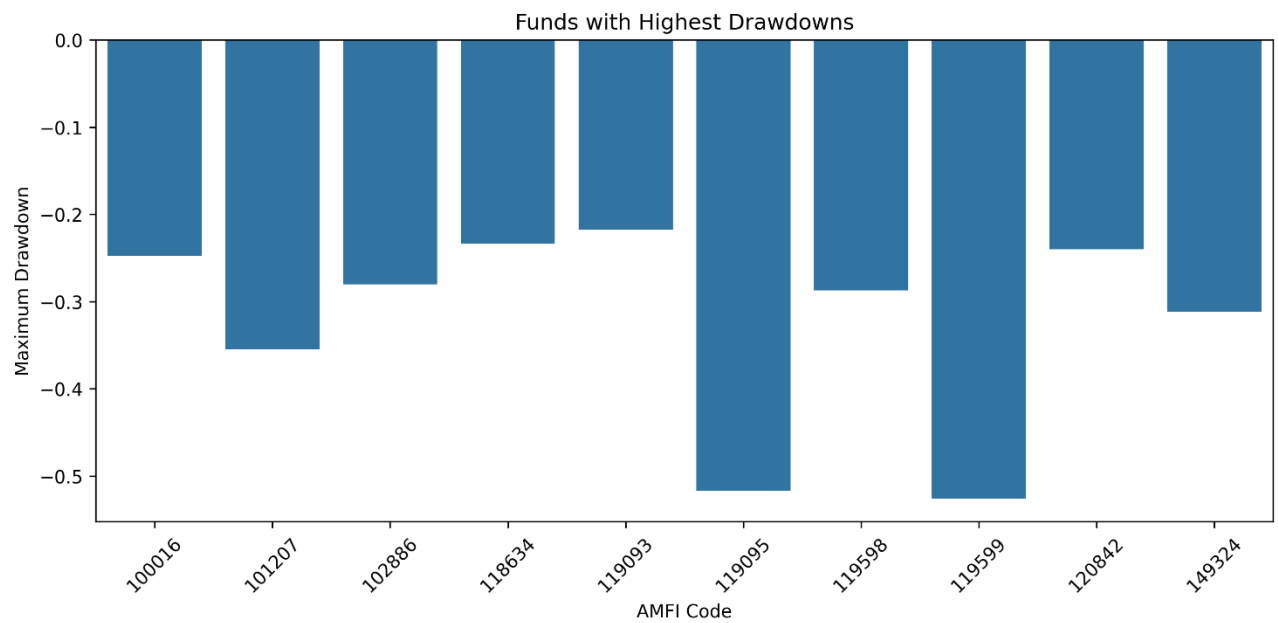
6.8 Maximum Drawdown Analysis

Maximum Drawdown measures the largest decline from a portfolio's peak value to its lowest point during a specified period. This metric reflects the worst historical loss experienced by an investor.

Drawdown analysis is particularly important because it highlights potential downside exposure during adverse market conditions.

The results revealed notable differences in downside risk among schemes. Some funds experienced substantial drawdowns during market corrections, while diversified portfolios maintained relatively lower losses.

Figure 18: Maximum Drawdown Across Mutual Fund Schemes



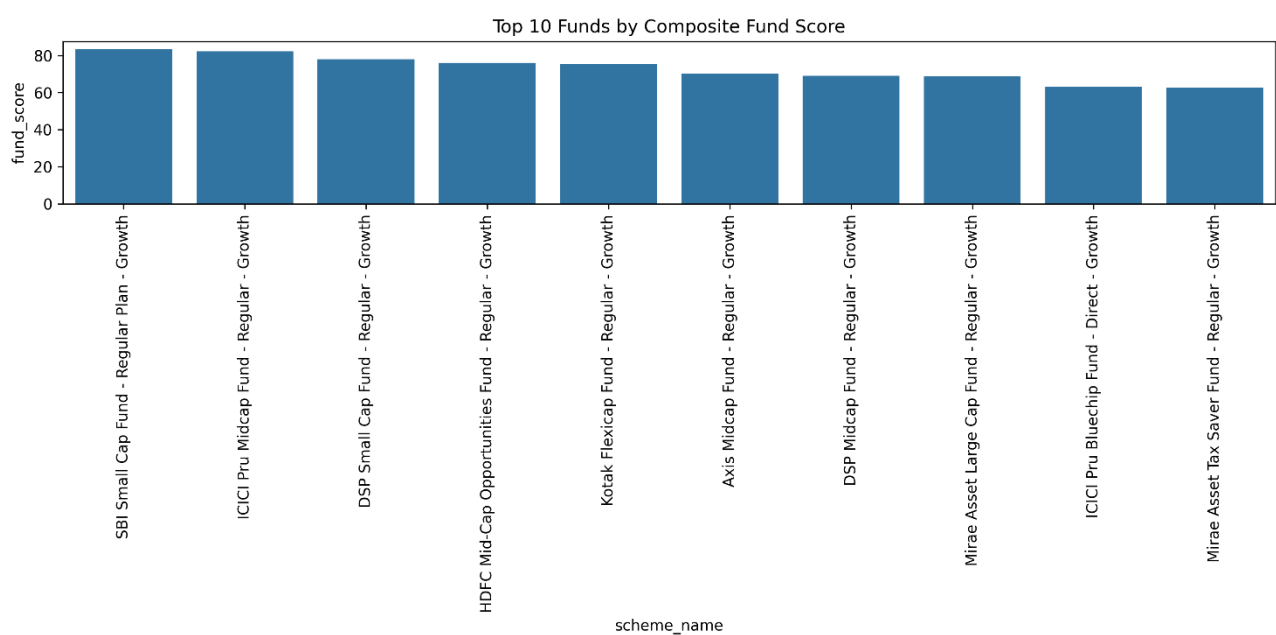
### Key Observations

- Maximum Drawdown varied significantly across schemes.
  - Diversified portfolios exhibited lower downside exposure.
  - Certain schemes experienced larger losses during market downturns.
  - Drawdown analysis provided valuable insights into capital preservation capabilities.
-

The performance analysis demonstrates that evaluating mutual funds requires consideration of both return generation and risk exposure. While certain schemes generated strong returns, risk-adjusted metrics such as Sharpe Ratio, Sortino Ratio, and Maximum Drawdown provided deeper insights into investment quality.

Overall, the analysis indicates that diversified and efficiently managed funds delivered superior long-term performance while maintaining acceptable levels of risk. The combination of return-based and risk-adjusted metrics provides investors with a comprehensive framework for evaluating mutual fund performance and making informed investment decisions.

Figure 19: Top Mutual Funds Based on Composite Fund Score



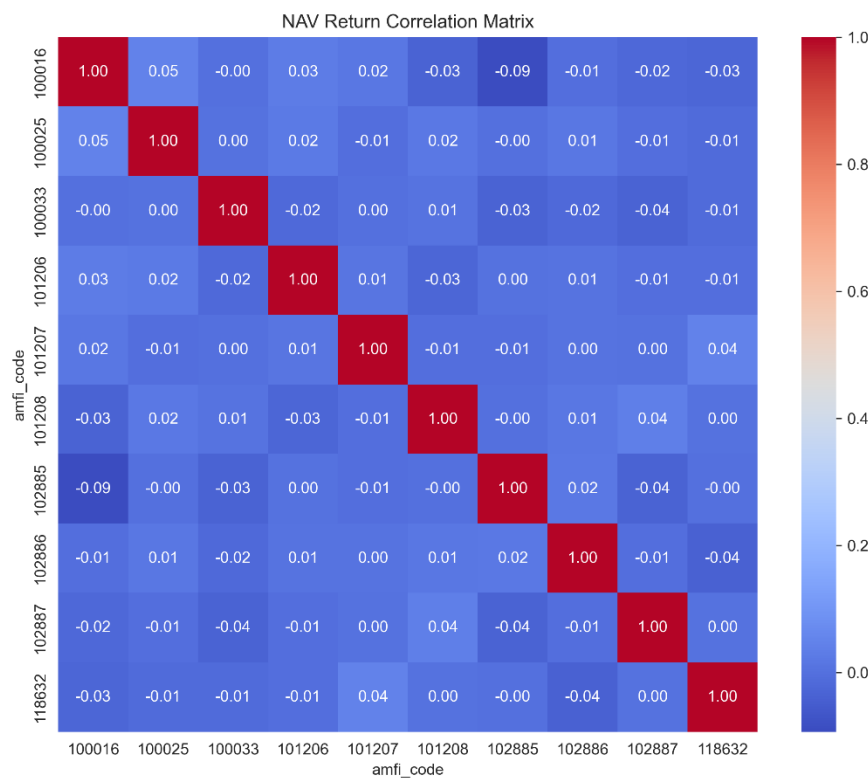
## 7. ADVANCED ANALYTICS AND RISK METRICS

### 7.1 Correlation Analysis

Correlation analysis was performed to evaluate the relationships among mutual fund returns and identify diversification opportunities within the investment universe. A correlation matrix was generated using historical NAV return data to measure the strength and direction of relationships between different mutual fund schemes.

The correlation heatmap provides a visual representation of how closely funds move together. Highly correlated funds tend to exhibit similar performance patterns, whereas lower correlations indicate diversification benefits.

Figure 20: Correlation Matrix of Mutual Fund Returns



#### Key Findings

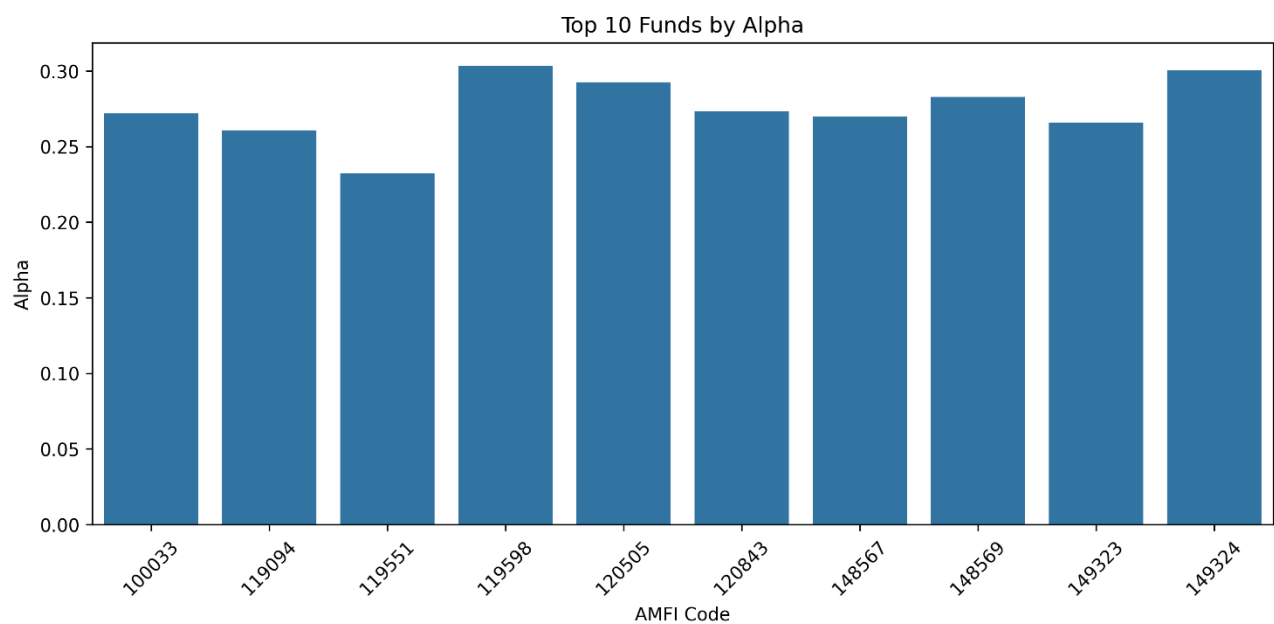
- Several equity-oriented schemes exhibited strong positive correlations.
- Highly correlated funds generally responded similarly to market movements.
- Lower correlation values highlighted diversification opportunities.
- Correlation analysis supports portfolio construction and risk management decisions.

## 7.2 Alpha Analysis

Alpha measures the excess return generated by a mutual fund relative to its benchmark after adjusting for market risk. A positive Alpha indicates that a fund has outperformed expectations based on its risk profile, while a negative Alpha suggests underperformance.

The Alpha analysis revealed differences in fund manager effectiveness and stock selection capabilities across schemes.

Figure 21: Top Mutual Funds Ranked by Alpha



### Key Findings

- Certain schemes generated positive Alpha, indicating benchmark outperformance.
- Higher Alpha values suggest effective active fund management.
- Alpha provides insight into the value added by fund managers.
- Positive Alpha contributes to superior long-term investment performance.

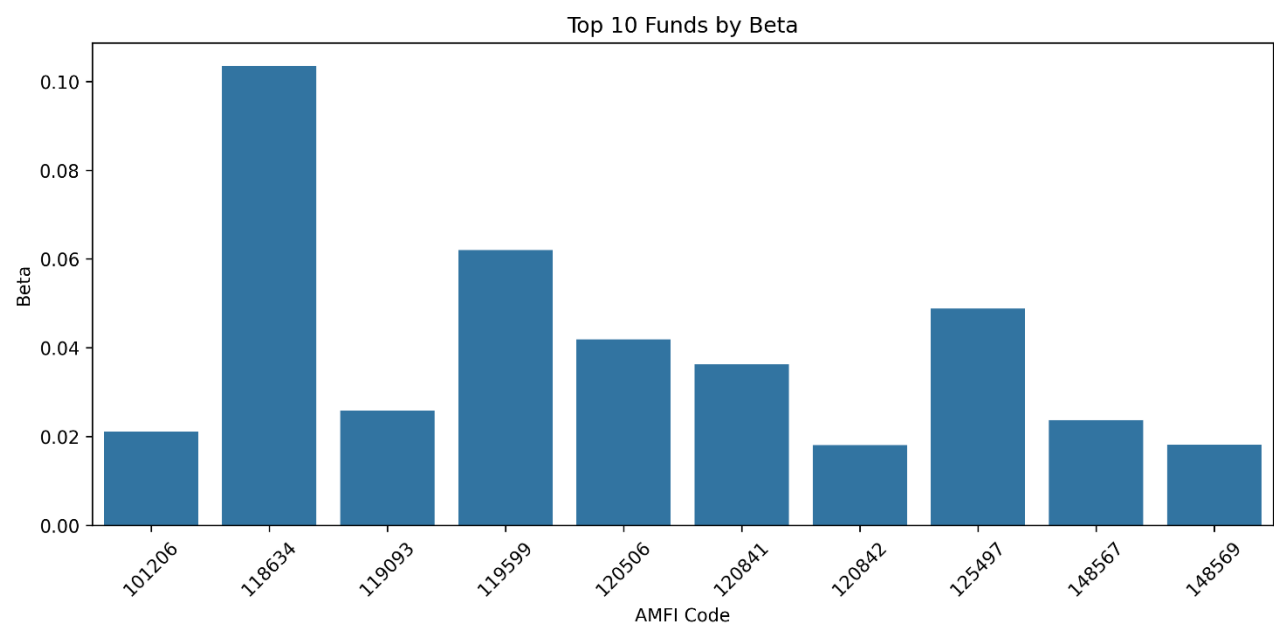
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## 7.3 Beta Analysis

Beta measures a fund's sensitivity to market movements. A Beta value greater than one indicates higher volatility relative to the benchmark, while a Beta value less than one indicates lower market sensitivity.

The Beta analysis helped assess market-related risk and understand how different schemes respond to changes in benchmark performance.

Figure 22: Top Mutual Funds Ranked by Beta



Key Findings

- High Beta funds exhibited greater responsiveness to market movements.
- Low Beta funds provided relatively stable performance during volatile periods.
- Beta is a useful measure of systematic risk.
- Investors can select funds based on desired risk exposure.

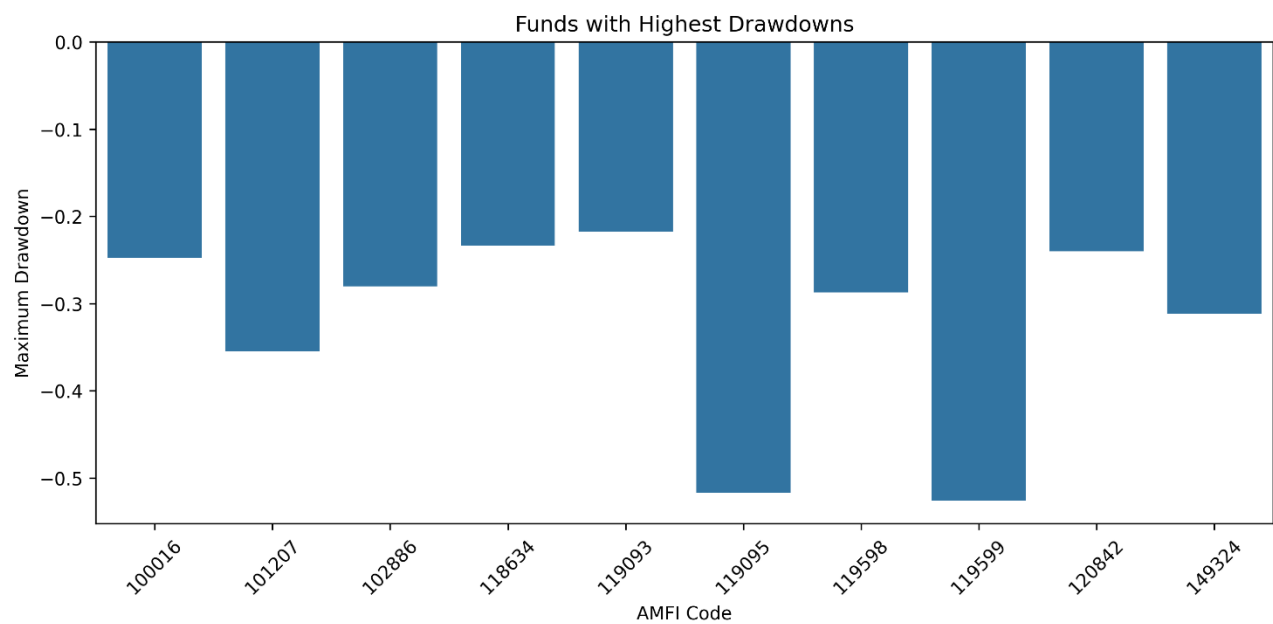
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7.4 Maximum Drawdown Analysis

Maximum Drawdown measures the largest decline from a portfolio's peak value to its lowest point during a specified period. This metric reflects the worst historical loss experienced by an investor and is a critical indicator of downside risk.

The analysis identified differences in downside exposure across mutual fund schemes.

Figure 23: Maximum Drawdown Comparison Across Mutual Funds



Key Findings

- Maximum Drawdown varied significantly across schemes.
- Some funds experienced larger losses during market corrections.
- Diversified portfolios generally exhibited lower drawdowns.
- Drawdown analysis is important for evaluating capital preservation capabilities.

7.5 Risk-Adjusted Performance Analysis

Risk-adjusted performance was evaluated using Sharpe Ratio and Sortino Ratio metrics. These measures assess how effectively funds generate returns relative to the risks undertaken.

Funds with higher Sharpe and Sortino Ratios demonstrated superior efficiency in converting risk exposure into investment returns.

Figure 24: Top Mutual Funds Ranked by Sharpe Ratio

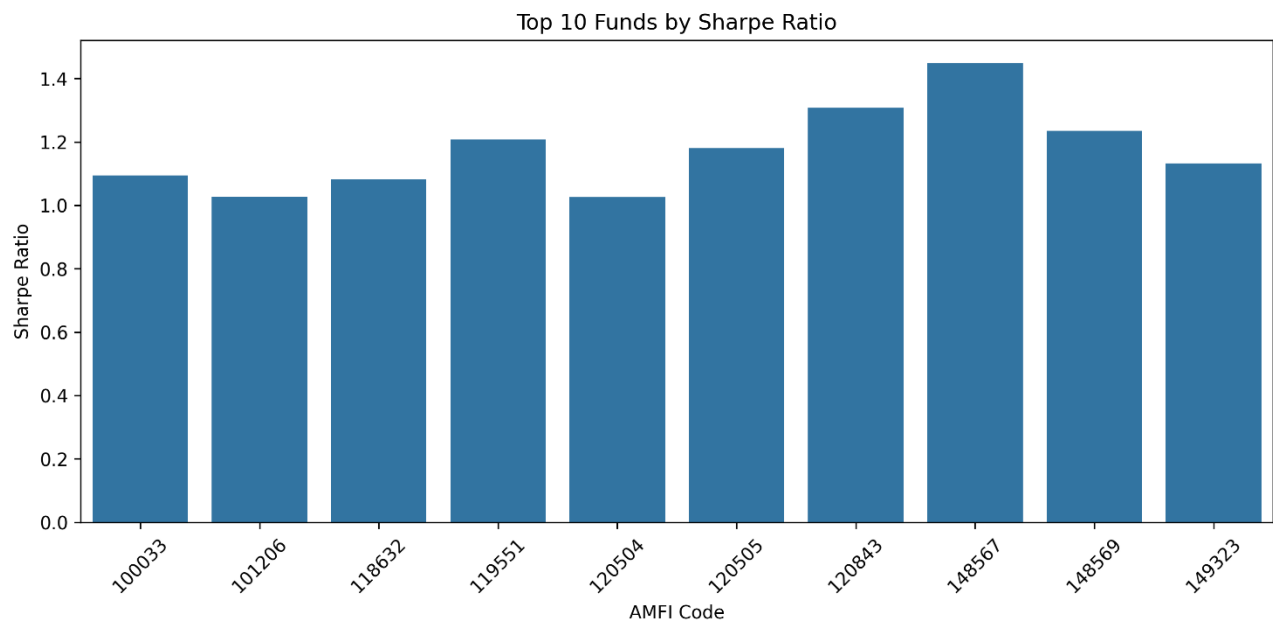
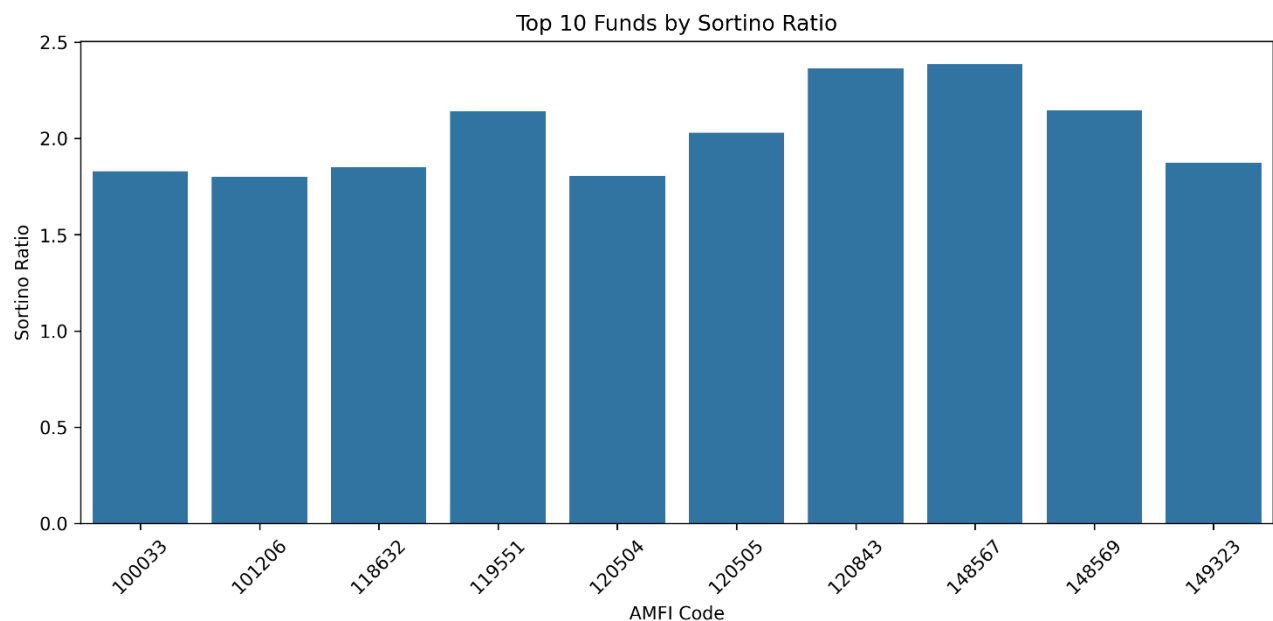


Figure 25: Top Mutual Funds Ranked by Sortino Ratio



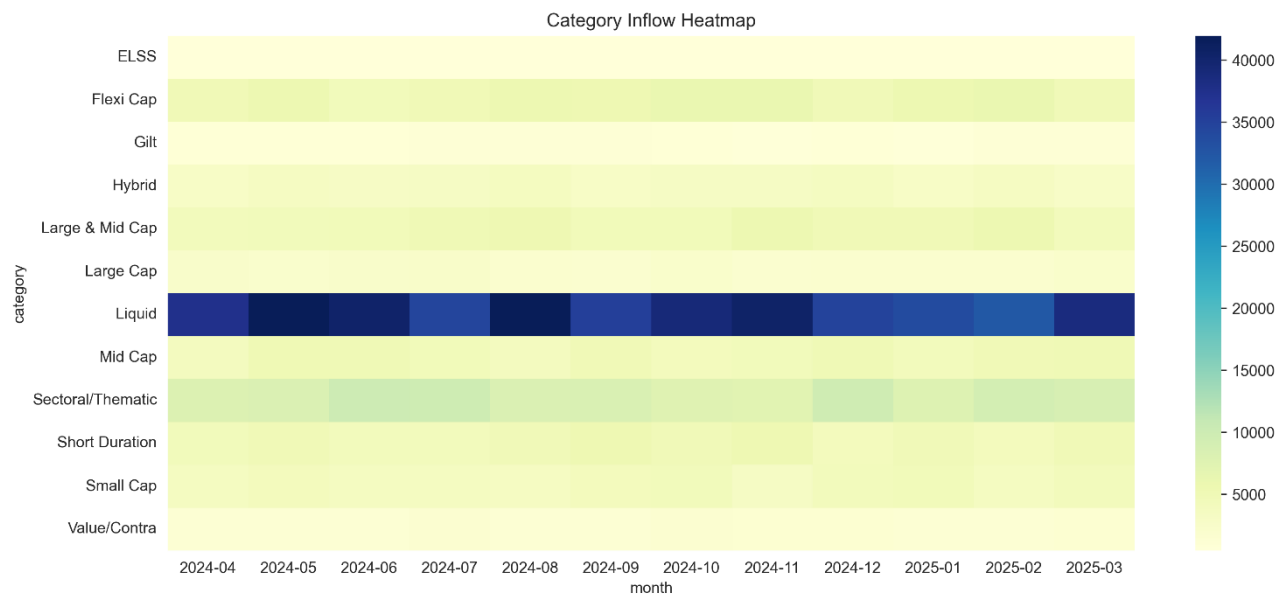
Key Findings

- Risk-adjusted performance differed significantly across schemes.
  - Some funds consistently delivered higher returns for each unit of risk.
  - Sortino Ratio provided additional insights into downside risk management.
  - Risk-adjusted metrics support more informed investment decisions.
-

7.6 Category Inflow Analysis

Category-wise inflow analysis was performed to identify investor preferences across mutual fund categories. The heatmap visualization highlights periods of strong inflows and outflows across different investment segments.

Figure 26: Category-wise Mutual Fund Inflow Heatmap



Key Findings

- Equity categories attracted significant investor inflows.
- Certain categories demonstrated seasonal investment patterns.
- Investor preferences shifted in response to market conditions.
- Category inflow analysis provides insights into market sentiment.

Risk Analytics Summary

The advanced analytics and risk assessment phase provided a deeper understanding of mutual fund behavior beyond traditional return measures. Correlation analysis highlighted diversification opportunities, Alpha and Beta metrics evaluated fund performance relative to market benchmarks, while Sharpe Ratio, Sortino Ratio, and Maximum Drawdown offered valuable insights into risk-adjusted performance and downside exposure.

These analyses collectively enable investors to evaluate not only how much return a fund generates, but also the quality and sustainability of those returns under varying market conditions.

## 8. DASHBOARD DEVELOPMENT AND VISUALIZATION

### 8.1 Dashboard Overview

A comprehensive interactive dashboard was developed using Microsoft Power BI to present the analytical insights generated throughout the Mutual Fund Analytics project. The dashboard serves as the final visualization layer of the analytics pipeline and enables users to explore mutual fund performance through an intuitive and interactive interface.

The dashboard integrates information from multiple datasets including NAV history, Assets Under Management (AUM), SIP inflows, investor demographics, portfolio holdings, and performance metrics. By consolidating these datasets into a single reporting environment, the dashboard provides stakeholders with a holistic view of mutual fund performance and investor behavior.

The primary objective of the dashboard is to transform complex financial data into actionable business insights that support investment evaluation and decision-making.

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### 8.2 Dashboard Features

The Power BI dashboard incorporates several interactive components designed to enhance user experience and analytical capabilities.

#### KPI Cards

Key Performance Indicator (KPI) cards provide quick access to important metrics such as total AUM, SIP inflows, number of investor folios, average returns, and risk measures.

#### NAV Trend Analysis

Interactive line charts allow users to visualize historical NAV movement and evaluate long-term fund growth patterns.

#### AUM Monitoring

AUM visualizations help users compare fund houses and assess market participation levels.

#### SIP Inflow Tracking

SIP inflow dashboards illustrate investor contribution trends and the growth of systematic investment behavior.

Investor Insights

Demographic and geographic visualizations help identify investor participation patterns across age groups, genders, and locations.

Risk Metrics

Risk-focused visuals present performance measures such as Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown.

Interactive Filters and Slicers

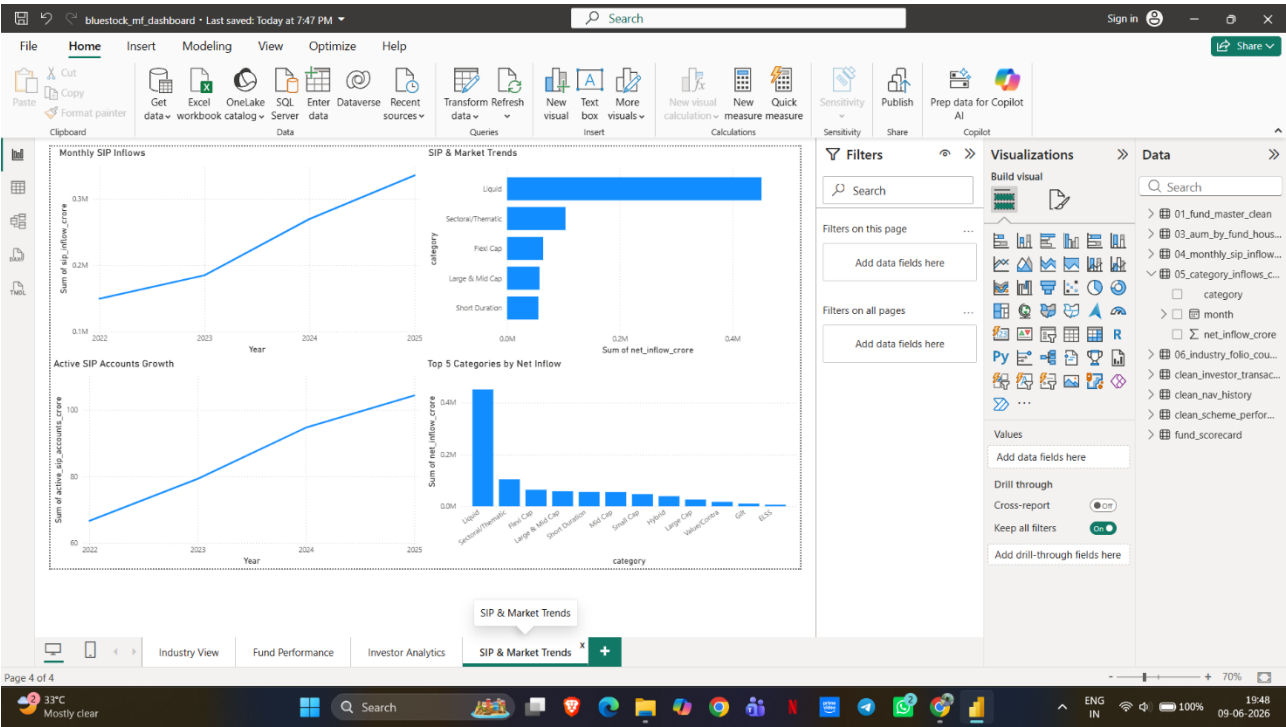
Users can dynamically filter data by scheme, fund category, time period, and investor segment, enabling personalized analysis and deeper exploration of the data.

8.3 Dashboard Screenshots

Dashboard Overview Page

The Overview Dashboard provides a high-level summary of the mutual fund ecosystem, including key metrics, growth indicators, and overall industry trends.

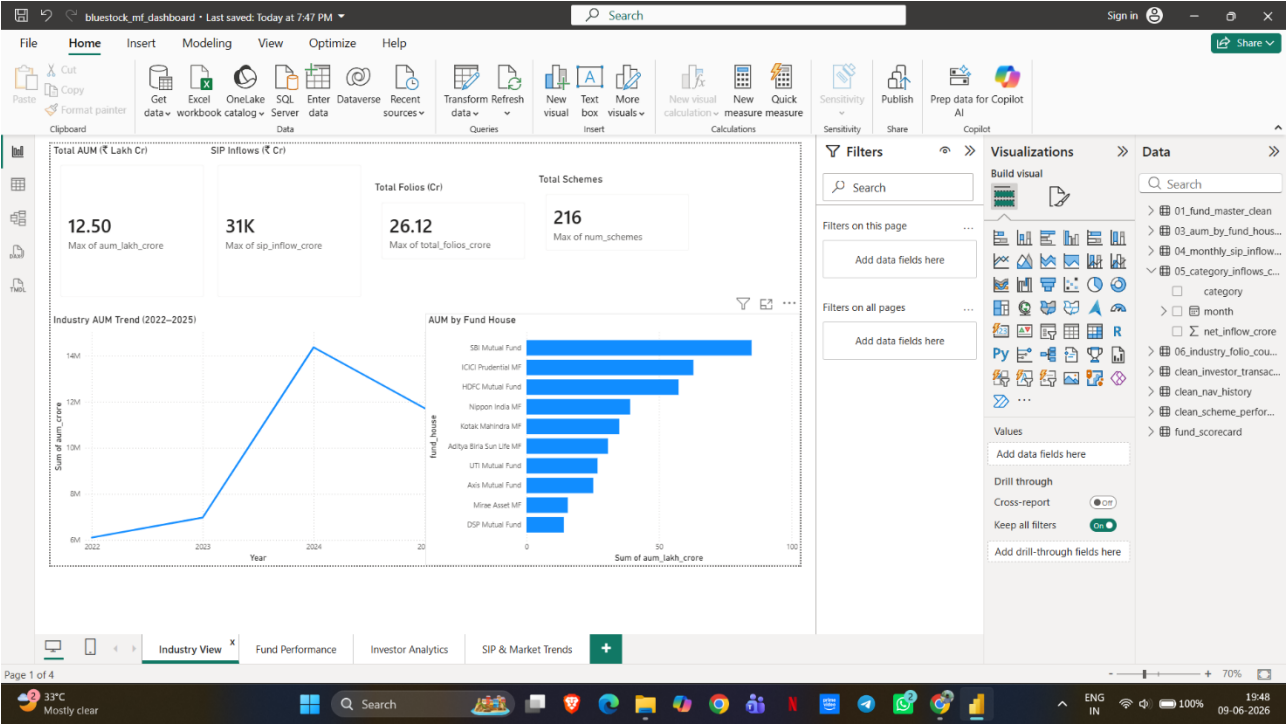
Figure 27: Industry Overview Dashboard



Performance Analytics Dashboard

The Performance Dashboard focuses on evaluating fund returns, CAGR, benchmark comparisons, and long-term investment growth.

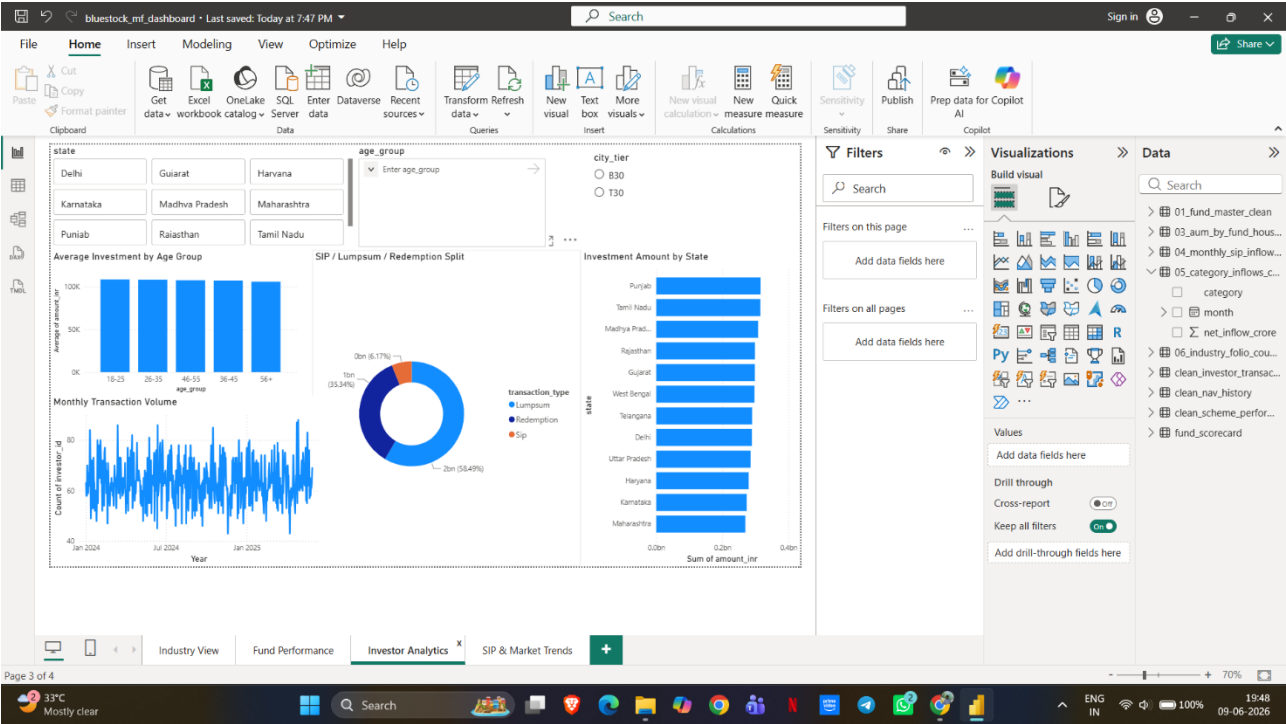
Figure 28: Fund Performance Dashboard



Fund Performance Dashboard

The Fund Performance Dashboard evaluates mutual fund schemes using return and risk metrics. The dashboard includes a Return vs Risk scatter plot, benchmark comparison visualization, and a fund score ranking table. These visualizations enable investors to compare schemes based on performance, volatility, and overall investment quality.

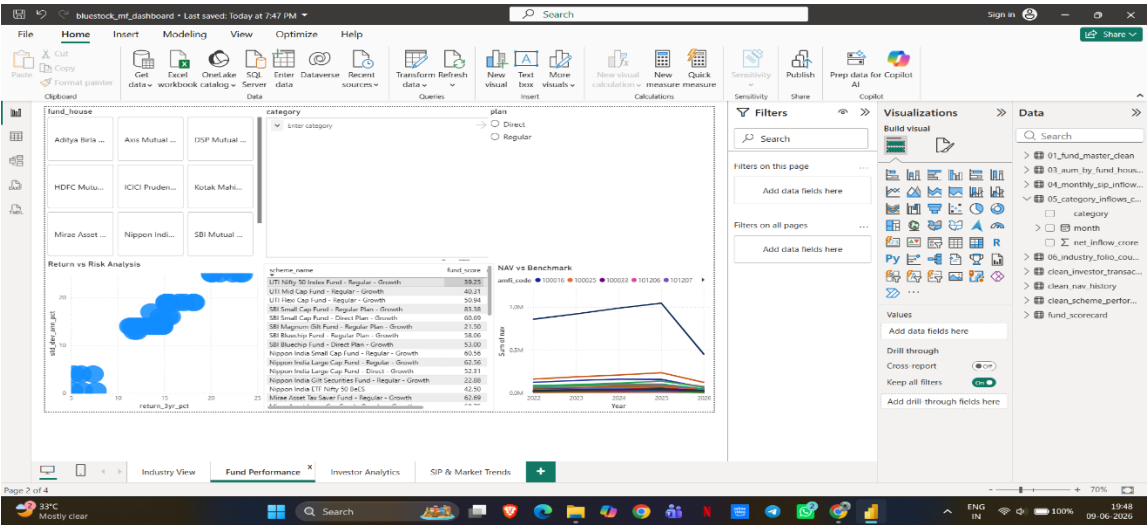
Figure 29: Investor Analytics Dashboard



Investor Insights Dashboard

The Investor Insights Dashboard highlights demographic participation, SIP behavior, folio growth, and geographic investment trends. These insights help understand investor preferences and market penetration.

Figure 30: SIP & Market Trends Dasboard



## **8.4 Business Benefits**

The Power BI dashboard provides significant business value by enabling data-driven decision-making and improving accessibility to analytical insights.

Key benefits include:

- Centralized view of mutual fund performance.
- Faster identification of investment opportunities.
- Enhanced risk assessment capabilities.
- Improved investor behavior analysis.
- Interactive exploration of financial data.
- Better support for strategic investment decisions.
- Simplified reporting and visualization.

The dashboard successfully bridges the gap between raw financial data and actionable business intelligence by presenting complex analytics in a visually intuitive and user-friendly format.

## 9. KEY FINDINGS

The analysis conducted throughout the Mutual Fund Analytics Capstone Project generated several meaningful insights regarding fund performance, investor behavior, portfolio composition, and market participation trends.

### Key Findings

1. Historical NAV analysis revealed a consistent long-term growth trend across most mutual fund schemes, demonstrating the effectiveness of long-term investing and the benefits of compounding.
2. Assets Under Management (AUM) increased steadily across major fund houses, indicating growing investor confidence and expanding participation in the mutual fund industry.
3. Monthly SIP inflows exhibited significant growth over the analysis period, highlighting increasing adoption of systematic investment plans as a preferred investment strategy among retail investors.
4. Active SIP account registrations increased consistently, reflecting growing financial awareness and long-term investment commitment among investors.
5. Industry folio counts showed continuous expansion, suggesting broader market penetration and increased participation from retail investors.
6. Equity-oriented mutual funds attracted substantially higher investor participation compared to debt-oriented schemes, indicating a preference for growth-focused investment products.
7. Investor demographic analysis revealed that the 26–35 age group contributed the highest investment volumes, making it one of the most active investor segments.
8. Geographic analysis demonstrated that T30 cities continue to account for the majority of mutual fund investments, although participation from B30 cities is steadily increasing.
9. Portfolio analysis indicated significant allocations toward Financial Services and Technology sectors, reflecting prevailing market trends and sector preferences.
10. Benchmark comparison showed that several mutual fund schemes delivered competitive performance relative to the NIFTY100 benchmark index.

11. Risk-adjusted performance metrics such as Sharpe Ratio and Sortino Ratio revealed significant variation across funds, highlighting differences in investment efficiency and risk management.
12. Correlation analysis identified diversification opportunities by revealing varying relationships among mutual fund returns.
13. Maximum Drawdown analysis demonstrated that diversified portfolios generally experienced lower downside risk during market corrections.
14. The Power BI dashboard successfully consolidated multiple analytical outputs into an interactive business intelligence platform, enabling efficient monitoring and decision-making.

Overall, the findings confirm that combining return analysis, risk evaluation, investor behavior studies, and visualization techniques provides a comprehensive framework for mutual fund assessment.

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## 10. LIMITATIONS

Although the project successfully achieved its objectives, several limitations were identified during implementation and analysis.

### Historical Data Dependency

The analysis was conducted using historical datasets. Past performance does not guarantee future results, and therefore investment decisions should not rely solely on historical observations.

### Absence of Real-Time Market Data

The project utilized static datasets rather than live market feeds. As a result, the dashboard reflects historical conditions and cannot provide real-time investment insights.

### Limited Forecasting Capability

The current implementation focuses on descriptive and diagnostic analytics. Predictive forecasting models were not incorporated within the project scope.

### Dataset Constraints

The analysis was limited to the datasets provided for the capstone project. Additional datasets such as macroeconomic indicators, interest rates, inflation measures, and market sentiment variables could further enhance analytical depth.

### Investor Behavior Variables

While demographic and transaction-level information was analyzed, more detailed behavioral indicators such as investor risk profiles and investment objectives were not available.

### Portfolio Optimization Scope

The project evaluates portfolio composition but does not implement advanced portfolio optimization techniques such as Modern Portfolio Theory or mean-variance optimization.

Despite these limitations, the project provides a strong analytical foundation and delivers valuable insights into mutual fund performance and investor participation trends.

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## 11. RECOMMENDATIONS

Based on the findings and limitations identified throughout the project, several recommendations are proposed for future enhancement and expansion.

### Real-Time Data Integration

Future versions of the platform should integrate live data feeds from AMFI, NSE, BSE, or financial APIs to provide up-to-date investment insights and dashboard reporting.

### Predictive Analytics and Forecasting

Machine learning models can be implemented to forecast NAV growth, SIP inflows, investor participation trends, and future fund performance.

### Portfolio Optimization Models

Advanced optimization techniques such as Efficient Frontier analysis, Mean-Variance Optimization, and Monte Carlo simulations can be incorporated to improve investment decision support.

### Personalized Recommendation Engine

A recommendation system can be developed to suggest suitable mutual funds based on investor goals, risk tolerance, investment horizon, and historical preferences.

### Automated Data Refresh Pipeline

The ETL process can be automated through scheduled workflows to ensure continuous data updates and minimal manual intervention.

### Advanced Investor Segmentation

Additional demographic and behavioral variables can be incorporated to improve investor segmentation and enable more targeted insights.

### Mobile and Cloud Deployment

The Power BI dashboard can be deployed through cloud platforms and optimized for mobile access, increasing accessibility and usability.

### Enhanced Risk Analytics

Future implementations can include Value at Risk (VaR), Conditional VaR, stress testing, and scenario analysis to provide deeper risk assessment capabilities.

## 12. CONCLUSION

The Mutual Fund Analytics Capstone Project successfully developed an end-to-end analytics platform for evaluating mutual fund performance, investor participation, and portfolio characteristics. The project combined data engineering, exploratory analysis, performance measurement, risk assessment, and business intelligence reporting into a unified analytical framework.

A structured ETL pipeline was implemented to transform raw mutual fund datasets into reliable and analysis-ready information. Exploratory Data Analysis provided valuable insights into NAV trends, AUM growth, SIP inflows, investor demographics, and portfolio composition. Performance metrics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown enabled comprehensive evaluation of both returns and risk.

The Power BI dashboard served as an interactive visualization layer, consolidating complex analytical outputs into intuitive reports that support investment evaluation and decision-making. Through dynamic charts, KPI cards, filters, and comparative analyses, the dashboard enhanced accessibility and usability of the analytical findings.

The project demonstrates the practical application of Python, Pandas, NumPy, Matplotlib, Seaborn, SQLite, and Power BI within the financial analytics domain. More importantly, it highlights the value of data-driven decision-making in mutual fund analysis and investment management.

Overall, the project achieved all intended objectives and established a scalable foundation for future enhancements involving real-time analytics, predictive modeling, portfolio optimization, and intelligent investment recommendation systems. The resulting platform provides investors, analysts, and financial professionals with a robust framework for understanding mutual fund performance and making informed investment decisions.

## 13. REFERENCES

1. Python Software Foundation. Python Documentation.
  2. Pandas Development Team. Pandas Documentation.
  3. NumPy Developers. NumPy Documentation.
  4. Matplotlib Documentation.
  5. Seaborn Documentation.
  6. SQLite Documentation.
  7. Microsoft Power BI Documentation.
  8. Association of Mutual Funds in India (AMFI).
  9. National Stock Exchange of India (NSE).
  10. Securities and Exchange Board of India (SEBI).
-

## APPENDIX

### Appendix A: Project Folder Structure

bluestock\_mf\_capstone/

├── data/

| ├── raw/

| ├── processed/

| └── db/

├── notebooks/

├── scripts/

├── reports/

├── dashboard/

├── sql/

└── README.md

---

### Appendix B: Technologies Used

- Python
  - Pandas
  - NumPy
  - Matplotlib
  - Seaborn
  - SQLite
  - Power BI
  - Git & GitHub
-

## **Appendix C: Generated Outputs**

### **Exploratory Data Analysis (EDA)**

- NAV Trend Analysis
- AUM Growth Analysis
- Monthly SIP Inflow Analysis
- Active SIP Accounts Growth Analysis
- Industry Folio Growth Analysis
- Equity vs Debt Folio Analysis
- Investor Demographic Analysis
- Geographic Participation Analysis
- Correlation Analysis

### **Performance Analytics**

- CAGR Analysis
- Sharpe Ratio Analysis
- Sortino Ratio Analysis
- Alpha-Beta Analysis
- Maximum Drawdown Analysis
- Fund Scorecard Analysis
- Benchmark Comparison Analysis

### **Advanced Analytics & Risk Metrics**

- Value at Risk (VaR) Analysis
- Conditional Value at Risk (CVaR) Analysis
- Rolling Sharpe Ratio Analysis
- Investor Cohort Analysis
- SIP Continuity Analysis

- Fund Recommendation Engine
- Sector HHI Concentration Analysis

### **Dashboard Outputs**

- Industry Overview Dashboard
  - Fund Performance Dashboard
  - Investor Analytics Dashboard
  - SIP & Market Trends Dashboard
- 

## **Appendix D: Git Version Control**

### **Day 1: Data Ingestion Complete**

**Commit ID:** a25db81

### **Day 2: Data Cleaning & SQLite Database Integration**

**Commit ID:** 9a47a11

### **Day 3: Exploratory Data Analysis (EDA)**

**Commit ID:** 6903d5e

### **Day 4: Performance Analytics**

**Commit ID:** f2d551b

### **Day 5: Power BI Dashboard Development**

**Commit ID:** f287aa5

### **Day 6: Advanced Analytics and Risk Metrics**

**Commit ID:** d6990b7

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## **Appendix E: GitHub Repository**

Repository URL:

[https://github.com/Pardhu9114/bluestock\\_mf\\_capstone](https://github.com/Pardhu9114/bluestock_mf_capstone)

The repository contains all project notebooks, scripts, processed datasets, dashboard files, reports, and version-controlled project deliverables developed throughout the capstone project.